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Conspicuous Gas, Cryptic Dust: Spectroscopy and Chromaticity of Complex Periodic Variables

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ABSTRACT

Complex periodic variables (CPVs) are young low-mass stars whose light curves show periodic dips indicative of transiting corotating material. The origin and composition of this material are unclear. Here we present new optical and near-infrared spectroscopy and photometry of four CPVs from Magellan, Keck, Hale, MuSCAT1, MuSCAT2, Tierras, KeplerCam, and TESS. The spectra imply that CPVs host magnetically-bound circumstellar plasma clumps, on the basis of sinusoidal-in-time Balmer emission out of transit, and Balmer dimming during transit. Yet large night-to-night changes in circumstellar hydrogen emissivity occur without clear changes in light curve morphology, suggesting that the sharp flux dips are caused not by circumstellar plasma but by dust. Optical chromaticities ($\delta \propto \lambda^{-\beta}$, β=0.79$\pm$0.16) support this, but the power law breaks in the near infrared, where a single power law under-predicts the depths observed at 2.1 μm. We therefore favor dips caused by dusty plasma clumps with opaque cores and optically thin halos, though we cannot rule out models in which the dust properties vary per-star or per-epoch. Observations at λ>2 μm and λ<0.4 μm would test this interpretation and clarify the dust's origin.

Keywords: Circumstellar matter (241), Stellar magnetic fields (1610), Stellar rotation (1629), Periodic variable stars (1213), Weak-line T Tauri stars (1795)

1. INTRODUCTION

Complex periodic variables (CPVs) present the kind of signal that astronomers usually consider easy: large amplitude, short period, and repeatable. Yet a decade after their discovery, the composition, structure, and origin of the occulting material remain unknown. CPVs are young M dwarfs, typically with rotation periods shorter than two days and, within current sensitivity limits, no disks (J. Stauffer et al. 2017; L. G. Bouma et al. 2024). The material trapped in their magnetospheres transits at a few stellar radii: the region occupied by the closest planets, and an environment that is otherwise hard to measure. While the impact of stellar photons on exoplanets is widely appreciated (K. France et al. 2016; J. E. Owen 2019), the particle environment can be just as influential (D. C. Catling & J. F. Kasting 2017), and means for studying it are scarce (R. O. P. Loyd et al. 2025). CPVs may provide such a means.

Figure 1 shows the main objects of interest in this article; over one hundred analogous systems have been reported by K2 and TESS (L. M. Rebull et al. 2016; J. Stauffer et al.

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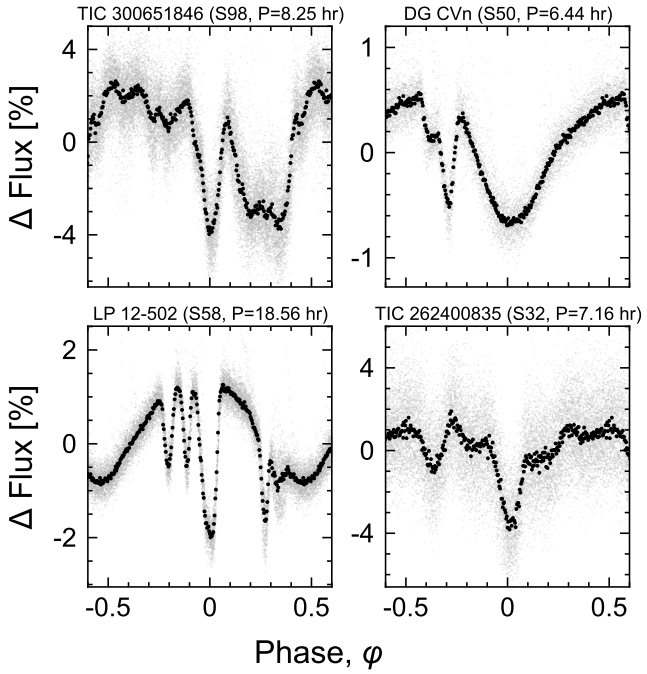


Figure 1. TESS light curves of the four targets. We acquired new spectra and multi-color photometry for TIC 300651846, DG CVn (TIC 368129164), LP 12-502 (TIC 402980664), and TIC 262400835. Panels show one month of TESS data folded on each object's period, with 120-second cadence flux measurements in gray and their average in black. For DG CVn, we removed a best–fit sinusoid to the P_2=2.60 hr secondary companion's rotation signal (3.1 AU projected separation; J.-L. Beuzit et al. 2004). The time evolution of each target from 2018–2025 is shown in the associated **online movies**: while the periods remain fixed, the morphologies evolve over months to years.

2017; L. M. Rebull et al. 2018; Z. Zhan et al. 2019a; L. M. Rebull et al. 2020; J. Stauffer et al. 2021; E. K. Palumbo et al. 2022; M. Popinchalk et al. 2023; L. G. Bouma et al. 2024). CPVs are identified from their structured, periodic optical light curves. About 3% of stars a few million years old show the behavior, declining to ≈0.3% by ≈110 Myr (L. M. Rebull et al. 2020). The leading interpretation is that clumps of charged circumstellar material corotate with the star (J. Stauffer et al. 2017). The physical requirement is a "centrifugal magnetosphere" (e.g. V. Petit et al. 2013): when the magnetospheric radius R_m exceeds the corotation radius R_c, the effective potential along closed field lines acquires local minima at radii beyond R_c, where the field enforces super-Keplerian motion and centrifugal support confines material against infall (R. H. D. Townsend & S. P. Owocki 2005). The trapped material is thus *in* corotation with the star, while residing *beyond* the corotation radius.

Spectroscopy offers a direct test of the corotating-material hypothesis, and the necessary data have only recently become available. Early spectroscopic characterization of CPVs established the class as weak-lined T Tauri stars (J. Stauffer et al. 2017) that were chromospherically active with no evidence for accretion. The first time-series spectroscopy of a CPV showed additional sinusoidal-in-time Hα emission at circumstellar velocities, tracing cool ($\lesssim 10^4$ K) plasma clumps at opposing stellar longitudes near 2-4 $R_\star$, consistent with a pinched torus of magnetically-trapped plasma (L. G. Bouma & M. M. Jardine 2025). Similar Balmer emission is seen among rapid magnetic rotators across a wide range of masses (J. F. Donati et al. 2000; R. H. D. Townsend et al. 2005; N. J. Dunstone et al. 2006; M. B. Skelly et al. 2008), as is a complementary dimming phenomenology (A. Collier Cameron & R. D. Robinson 1989; A. Collier Cameron & J. A. Woods 1992; T. Q. Cang et al. 2020). Yet those stars do not show complex photometric variability, and before this work, TIC 141146667 was the only unambiguous CPV reported to show both, leaving the question of universality open. Do all CPVs harbor bound plasma? Which of the observed morphological features, such as opposing clumps, night-to-night variability, or Balmer dimming during transits, generalize across the class?

While spectroscopy can reveal the gas, multicolor photometry provides independent evidence for what causes the dips. Ground-based campaigns have established that CPV dips deepen toward the blue, with power-law slopes $\beta \approx 0$–1.5 for $\delta \propto \lambda^{-\beta}$ (M. Onitsuka et al. 2017; Y. Tanimoto et al. 2020; M. N. Günther et al. 2022; C. Koen 2023). This behavior suggests extinction by sub-micron dust. Three issues, however, complicate the interpretation. First, published dip depths rest on heterogeneous detrending choices, which propagate into the inferred chromaticity. Second, if the occulting material is partially opaque, the observed slope is diluted relative to the intrinsic opacity law, an effect that has so far been ignored. Third, the models considered have rarely extended beyond single power laws. The exception is the pioneering study of PTFO 8-8695 by Y. Tanimoto et al. (2020), who explored broken extinction laws, albeit interpreting that system's CPV (L. G. Bouma et al. 2020) as a planet candidate. These limitations matter because the chromaticity directly links to the question of where the dust in these systems comes from. Are these M dwarfs condensing out dust within their own magnetospheres, or trapping the detritus of planet formation? Neither conclusion can be drawn without first establishing what the occulting material does to starlight.

In this work we present time-series spectroscopy and multicolor photometry of four CPVs (Figure 1): 18 nights of ground-based spectroscopy and nine nights of ground-based photometry acquired between 2020 and 2025, usually with simultaneous TESS coverage. High-resolution spectroscopy at sensitivity comparable to the TIC 141146667 observations yields a detection of sinusoidal circumstellar Balmer emission for a second CPV, TIC 300651846, and upper limits for LP 12-502. Lower-resolution spectra with simultaneous photometry show that Balmer dimming during broadband flux dips is ubiquitous, though often with timing offsets. For the chromaticity, we homogeneously analyze our new photometry together with the archival literature, placing 51 dip depths across five stars on a common scale, and we compare four

extinction-law models. Under the assumption that a universal extinction law applies across the population, a broken power law between the optical and near-infrared is preferred; absent that assumption, models allowing star-to-star or epoch-to-epoch variability fit as well or better.

Section 2 describes the observations and reductions; Section 3 presents the interpretation of the spectroscopy and the extinction-law fits; Section 4 discusses the structure and possible origins of the occulting material; Section 5 concludes.

2. NEW OBSERVATIONS

2.1. *Selection Function*

We selected targets from the known CPV census weighing brightness, period, morphology, accessibility, and binarity (L. M. Rebull et al. 2016; J. Stauffer et al. 2017; L. M. Rebull et al. 2018; Z. Zhan et al. 2019a; L. M. Rebull et al. 2020; J. Stauffer et al. 2021; E. K. Palumbo et al. 2022; M. Popinchalk et al. 2023; L. G. Bouma et al. 2024). Brighter stars enable higher S/N. Shorter periods maximize the fraction of a cycle covered per night, and hence the probability of catching a sharp dip, since morphologies are unpredictable beyond a few weeks. Sharp, large-amplitude dips are most easily distinguished from spot-induced variability. Proximity to the TESS continuous viewing zones maximizes the number of schedulable windows for simultaneous photometry and spectroscopy. Finally, we were agnostic to binarity, since no CPVs with close ($<$0.1 AU) companions capable of affecting the circumstellar environment are known.

These considerations yielded five targets. LP 12-502, DG CVn, TIC 300651846, and TIC 262400835 have new data presented here; TIC 141146667 was discussed in L. G. Bouma & M. M. Jardine (2025). Table 1 lists the stellar properties, and Tables 2 and 3 the observations.

This work is the first to highlight TIC 262400835 (Gaia DR3 3234917110482839680) as a CPV. It was found in 2020 through CDIPS (L. G. Bouma et al. 2019) but omitted from the L. G. Bouma et al. (2024) catalog, whose automated methods did not recover it. Its short period and complex morphology motivated TESS DDT028, during which we also acquired MuSCAT1 and MuSCAT2 photometry. R. M. P. Kerr et al. (2021) placed it in the $\approx$19 Myr "Greater Taurus Subgroup 3A". The system is likely a triple. A resolved companion, TIC 262400836 (Gaia DR3 3234920065420339712), lies 10.″6 west ($\sim$1,800 AU projected) with consistent parallax and proper motion. The companion is 0.3 magnitudes bluer in BP-RP at near-identical G, and so is likely the primary. TIC 262400835 itself is likely an unresolved pair, based on its Gaia DR3 RUWE (1.5) and photometric excess. A single 21″ TESS pixel shows three periods: the complex signal at P_1=7.16 hr, and sinusoidal signals at P_2=4.31 day and P_3=12.70 hr. The MuSCAT data (Section 2.8) localize the 7.16 hr signal to TIC 262400835, which suffices for this work. A. N. Heinze et al. (2018) reported the resolved companion (TIC 262400836) as a P=4.3 day close binary; we recover the period but see no eclipses in the TESS data.

2.2. *Magellan/MagE*

We observed TIC 300651846 (V=14.1) over four consecutive nights spanning 18 January to 21 January 2025 using the Magellan Echellette (MagE; J. L. Marshall et al. 2008) spectrograph on the Baade 6.5 m Magellan telescope. We used the 1″$\times$10″ slit ($R\approx$4,100, $\delta v\approx$73 km s^{-1}), with a 12 minute cadence, and aligned the slit along the parallactic angle while taking thorium-argon arcs for wavelength calibration once per hour. The seeing averaged 0.″8 over all four nights, rarely exceeding 1″. We extracted 1-D spectra from the resulting 161 science-ready echellograms using the CarPy data reduction package (D. D. Kelson et al. 2000; D. Kelson 2003), with S/N$\approx$160 per resolution element in the continuum near Hα. We timed these observations to be simultaneous with TESS Sector 88. Adopting the stellar radius and period from L. G. Bouma et al. (2024), TIC 300651846 has $v_{\rm eq}$=89.7 km s^{-1}, implying near-critical sampling of the line core. Figure 2 shows the resulting time-dynamic spectra in the vicinity of Hα, where the bottom row subtracts the time-averaged spectrum from each night to highlight variability—whether in the line core, or in the circumstellar environment.

2.3. *Magellan/MIKE*

We additionally observed TIC 300651846 on the night of 2 January 2025 using the Magellan Inamori Kyocera Echelle (MIKE; R. Bernstein et al. 2003) spectrograph on the Clay 6.5 m Magellan telescope. We used the 0.7″$\times$5″ slit, delivering $R\approx$35,000 ($\delta v\approx$9 km s^{-1}) in the red arm and $R\approx$28,000 ($\delta v\approx$11 km s^{-1}) in the blue, and opted to read out unbinned in the spectral direction, and 2$\times$ binned in the spatial direction. We bracketed every science observation with a thorium-argon arc, since the main expected advantage of this observation configuration relative to MagE was the velocity resolution, which oversamples the line core. The seeing over the night averaged 0.″9. We extracted 1-D spectra from the resulting 23 science-ready echellograms using CarPy (D. D. Kelson et al. 2000; D. Kelson 2003), yielding S/N$\approx$54 per resolution element in the Hα continuum. These observations were executed simultaneously with TESS Sector 87. Figure 3 shows the resulting time-dynamic spectra.

2.4. *Hale/DBSP*

We used the Double Spectrograph (DBSP; J. B. Oke & J. E. Gunn 1982) on the Palomar 200-inch telescope (Hale/P200) to observe LP 12-502 over seven nights: 11 November to 12 November 2023, 7 December 2023, and 4 November to 8 November 2024. We additionally observed DG CVn (V=12.0), a known unresolved binary (R. A. Osten et al. 2016) in which one component is a CPV, over the night of 28 April 2024. We used the 1.″5 slit, with the 1200 lines/mm grating, blazed at 5000 Å on the blue arm, and 1200 lines/mm, blazed at 7100 Å on the red arm. This delivered $R\approx$2,200 at Hα, corresponding to $\delta v\approx$140 km s^{-1}. For LP 12-502, $v_{\rm eq}$=24.2 km s^{-1}, yielding an experiment focused on equivalent width measurement, rather than velocity resolution. For DG CVn ($v_{\rm eq}$=136 km s^{-1}) the velocity

Table 1. Overview of Stellar Properties

Star	P (hr)	$T_{\rm eff}$ (K)	$R_\star$ ($R_\odot$)	$M_\star$ ($M_\odot$)	Age (Myr)	$v_{\rm eq}$ (km s^{-1})	$R_{\rm c}$ ($R_\star$)	Multiple?
TIC 300651846	8.25	3159^{+56}_{-84}	$0.61^{+0.04}_{-0.03}$	$0.40^{+0.02}_{-0.03}$	23	89.7 ± 5.2	2.49 ± 0.15	N
LP 12-502	18.56	3090^{+31}_{-59}	$0.37^{+0.02}_{-0.01}$	$0.22^{+0.01}_{-0.02}$	80	24.2 ± 1.0	5.78 ± 0.27	N
DG CVn	6.44	3127^{+51}_{-42}	$0.72^{+0.02}_{-0.03}$	$0.40^{+0.02}_{-0.06}$	150	135.8 ± 4.7	1.79 ± 0.09	Y
TIC 262400835	7.16	3074^{+64}_{-58}	$1.00^{+0.04}_{-0.05}$	$0.30^{+0.03}_{-0.04}$	19	169.6 ± 7.6	1.26 ± 0.07	Y

NOTE—DG CVn is TIC 368129164; LP 12-502 is TIC 402980664. Following L. G. Bouma et al. (2024), effective temperatures and radii are from SED fits and masses are from PARSEC isochrones (Y. Chen et al. 2014). This procedure may yield biased results for binaries unresolved in the underlying photometry (*i.e.*, for the latter two systems). The age for LP 12-502 comes from co-moving pre-main sequence stars in its vicinity (B. M. Tofflemire et al. 2021); ages for the remaining stars are from association memberships (J. Gagné et al. 2026) and have relative uncertainties $\approx \pm 15\%$. The equatorial velocity $v_{\rm eq}=2\pi R_\star/P$ and the Keplerian corotation radius $R_{\rm c}=(GM_\star P^2/4\pi^2)^{1/3}$ are derived quantities. The multiplicity column is Yes/No: DG CVn is a resolved binary (3.1 AU projected separation; J.-L. Beuzit et al. 2004) in which one component is the CPV; TIC 262400835 is likely an unresolved pair within a hierarchical triple (see text).

Table 2. Spectroscopic Observations

Star	Spectrograph	Photometer	Night (UT)	Gas Resolvable?	Gas Emission?	Hα Dips?	Flare Var.?
LP 12-502	DBSP	Tierras	2023 Nov 11	×	×	✓	✓
LP 12-502	DBSP	Tierras	2023 Nov 12	×	×	✓	✓
LP 12-502	DBSP	×	2023 Dec 07	×	×	–	–
LP 12-502	DBSP	TESS	2024 Nov 04	×	×	✓	✓
LP 12-502	DBSP	TESS	2024 Nov 05	×	×	×	✓
LP 12-502	DBSP	TESS	2024 Nov 06	×	×	×	✓
LP 12-502	DBSP	TESS	2024 Nov 07	×	×	✓	✓
LP 12-502	DBSP	TESS	2024 Nov 08	×	×	×	✓
LP 12-502	HIRES	×	2023 Nov 23	✓	×	–	–
LP 12-502	HIRES	TESS	2024 Dec 03	✓	×	✓	✓
LP 12-502	HIRES	TESS	2024 Dec 10	✓	×	×	✓
DG CVn	DBSP	Tierras	2024 Apr 28	?	×	✓	×
TIC 300651846	MIKE	TESS	2025 Jan 02	✓	✓	✓	×
TIC 300651846	MagE	TESS	2025 Jan 18	✓	✓	✓	?
TIC 300651846	MagE	TESS	2025 Jan 19	✓	✓	✓	×
TIC 300651846	MagE	TESS	2025 Jan 20	✓	×	✓	×
TIC 300651846	MagE	TESS	2025 Jan 21	✓	×	✓	×
TIC 141146667[†]	HIRES	TESS	2024 Feb 17	✓	✓	✓	?

NOTE— The last four columns indicate (i) whether the spectrograph had sufficient resolution relative to the stellar equatorial velocity to resolve bound circumstellar gas in emission, (ii) whether such gas was seen, (iii) whether the Hα equivalent width time series showed dips synchronous with broadband flux dips, and (iv) whether the spectra and photometry showed synchronous evidence for flare-induced variability. [†]Data for TIC 141146667 were described in L. G. Bouma & M. M. Jardine (2025).

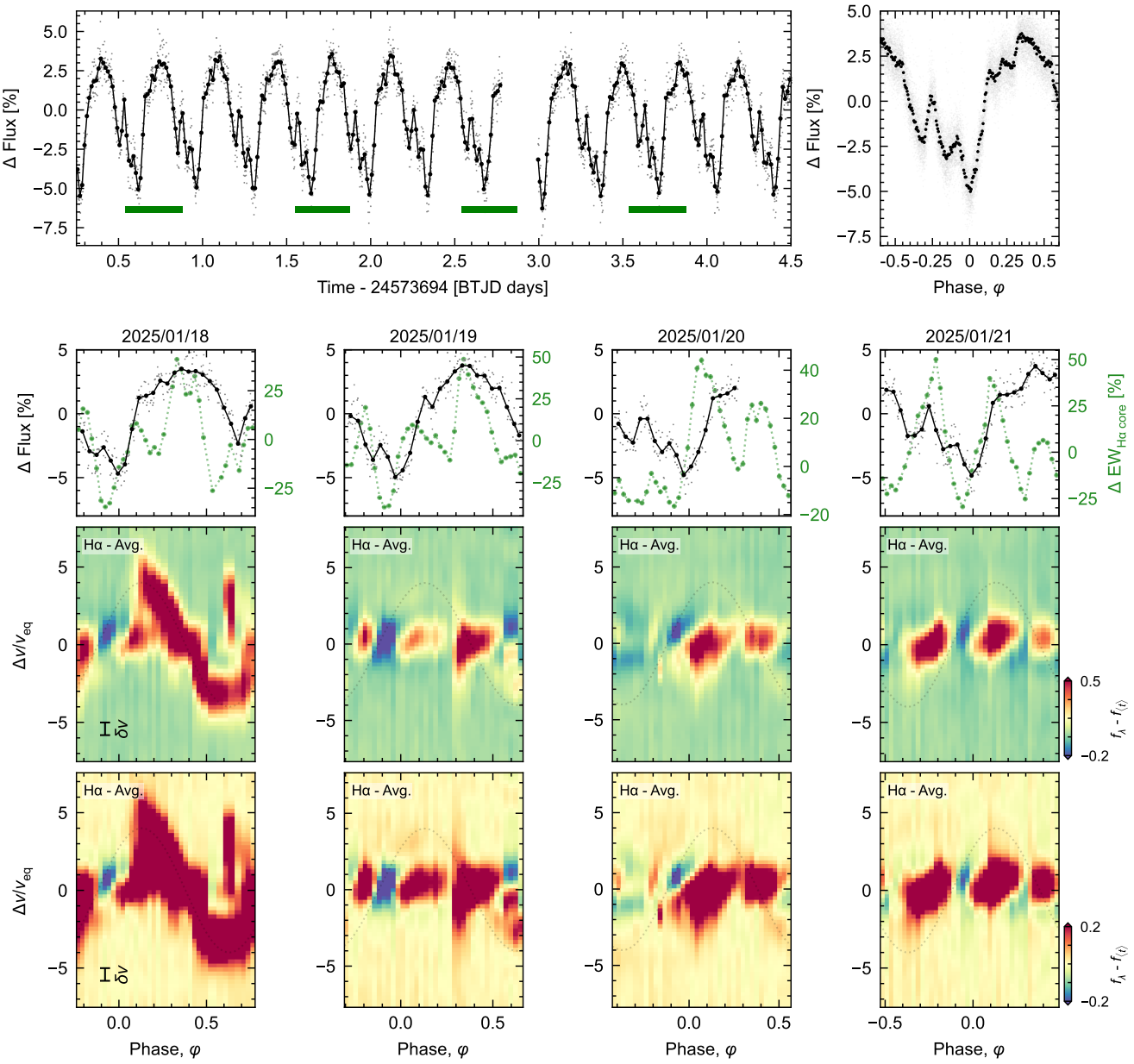


Figure 2. Circumstellar Balmer emission varies while the light curve does not. Data are from four successive nights on Magellan/MagE for TIC 300651846 (P = 8.25 hr). The top-left panel shows TESS data (raw in gray; 20 minute time binning in black); the next row zooms in on times when MagE and TESS overlap, and shows the line-core Hα equivalent width (at $|v/v_{\rm eq}| < 1$) with green markers. The lower two rows show time-series Hα profiles at two different color stretches, with the time-averaged mean profile subtracted. The average TESS light curve from 14 January to 11 February 2025 is in the top right. Large amplitude sinusoidal-in-time Balmer emission is visible on 18 January 2025. Such emission is weakly visible from ϕ=0.3-0.6 on 19 January and 20 January, and not present on 21 January. Despite the spectroscopic changes, the complex morphology in the TESS light curves remained stable. The error bar in the lower left indicates the velocity resolution. The faint model is a sinusoid with K=4 $v_{\rm eq}$ and ϕ_0=-0.12.

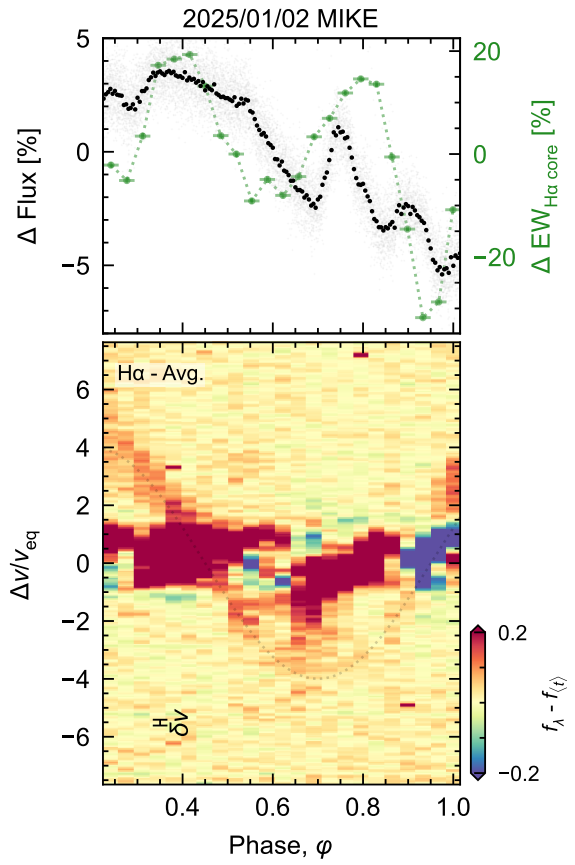


Figure 3. Magellan/MIKE Hα time series of TIC 300651846 on 2 January 2025. These data have eight times higher resolution than the MagE data in Figure 2; we bin by a factor of two in velocity for clarity, and again phase to $P = 8.25$ hr. The circumstellar emission mirrors that seen on 19 and 20 January. Sinusoidal-in–time Hα emits at ≈10% of the continuum. Absorption in the line core from ϕ=0.9-1.0 occurs during the transit of the corresponding clump of plasma. The ephemeris is the same as in Figure 2; the average Hα line profile (not shown) is similarly in emission with EW=6.0 ± 0.3 Å. The faint model is a sinusoid with K=4 $v_{\rm eq}$ and ϕ_0=-0.05, slightly offset from Figure 2.

sampling near the line core was near-critical. For LP 12-502, we used 10 minute exposures, yielding 442 spectra with 73.7 total hours of open-shutter time, typically with S/N≈200 per resolution element. For DG CVn, we used six minute exposures, yielding 49 spectra over a 4.3 hour baseline. We performed the extraction using `DBSP_DRP` (M. S. Mandigo-Stoba et al. 2022a), which is an extension of the `PypeIt` suite (J. X. Prochaska et al. 2020). We then measured the equivalent widths of the highest S/N Balmer lines—Hα (n=3→2), Hβ (n=4→2), and Hγ (n=5→2)—by numerically integrating the resulting continuum-normalized spectrum using a one-component Gaussian with free amplitude, width, and mean. We estimated the uncertainties on each line width through a Monte Carlo procedure that bootstrapped against the local scatter in the continuum.

The two LP 12-502 nights in November 2023 were conducted simultaneously with the Tierras photometer at FLWO (J. García-Mejía et al. 2020), detailed in Section 2.10 below. The remaining observations, save for 7 December 2023, were conducted simultaneously with TESS. Figure 4 summarizes the results from the most informative nights: 11 November to 12 November 2023, 4 November and 7 November 2024 for LP 12-502, and 28 April 2024 for DG CVn. Figure 5 provides further detail on the Hα, Hβ, and Hγ equivalent widths and line profiles for the November 2023 nights. Data from the remaining nights of 5 November and 6 November 2024 for LP 12-502 are presented in Appendix A—these observations covered orbital phases with no "complex" flux dips; only a single broadband flare on 5 November yielded a corresponding spike in the Balmer lines similar to those visible in Figure 4. The night of 7 December 2023 yielded viable DBSP data, but a TESS perigee contact extended into the DBSP observation, leaving insufficient overlapping photometry for further analysis.

2.5. *Keck/HIRES*

We observed LP 12-502 (V=13.9) over 23 November 2023, 3 December 2024, and 10 December 2024 using the High Resolution Echelle Spectrometer (HIRES; S. S. Vogt et al. 1994) on the 10 m Keck I telescope. We used the C2 decker (0.″86×14.″0) in the red configuration, yielding a resolution R≈45,000 (δv≈6.7 km s^{-1}). The seeing averaged 1.″2, 1.″6, and 2.″1 on the three nights, respectively. With 15-20 minute cadence, this configuration delivered an S/N of 70-120 per resolution element in the Hα order's continuum, and a total of 20.1 hours of open-shutter time. The star's radius and period from L. G. Bouma et al. (2024) imply an equatorial velocity ($2\pi R_\star/P$) of $v_{\rm eq}$=24.2 km s^{-1}, so the instrument provided a few resolution elements across the line core. We reduced the echellogram using the standard techniques of the California Planet Survey (A. W. Howard et al. 2010).

We acquired simultaneous photometry for the December 2024 nights from TESS. We attempted to acquire simultaneous Tierras data for the November 2023 night but had poor weather in Arizona. Appendix A details the results. In short, while the HIRES spectra and photometry of LP 12-502 presented abundant evidence for flare-induced variability, we observed only a ≈one hour partial transit of a single "complex" dip, which was coincident with excess absorption in Balmer line cores.

2.6. *Hale/TripleSpec*

We observed LP 12-502 on 27 January 2024 (UT) using the Triple Spectrograph (TripleSpec) mounted at the Cassegrain focus of the Palomar 200-inch (5.1 m) telescope. TripleSpec is a medium-resolution, cross-dispersed near-infrared spectrograph that provides simultaneous spectral coverage from 1.0–2.4 μm across five spectral orders (T. L. Herter et al. 2008). The instrument uses a 1″ × 30″ slit and provides a resolving power of $R \sim 2600$, with approximately 2.7 pixels

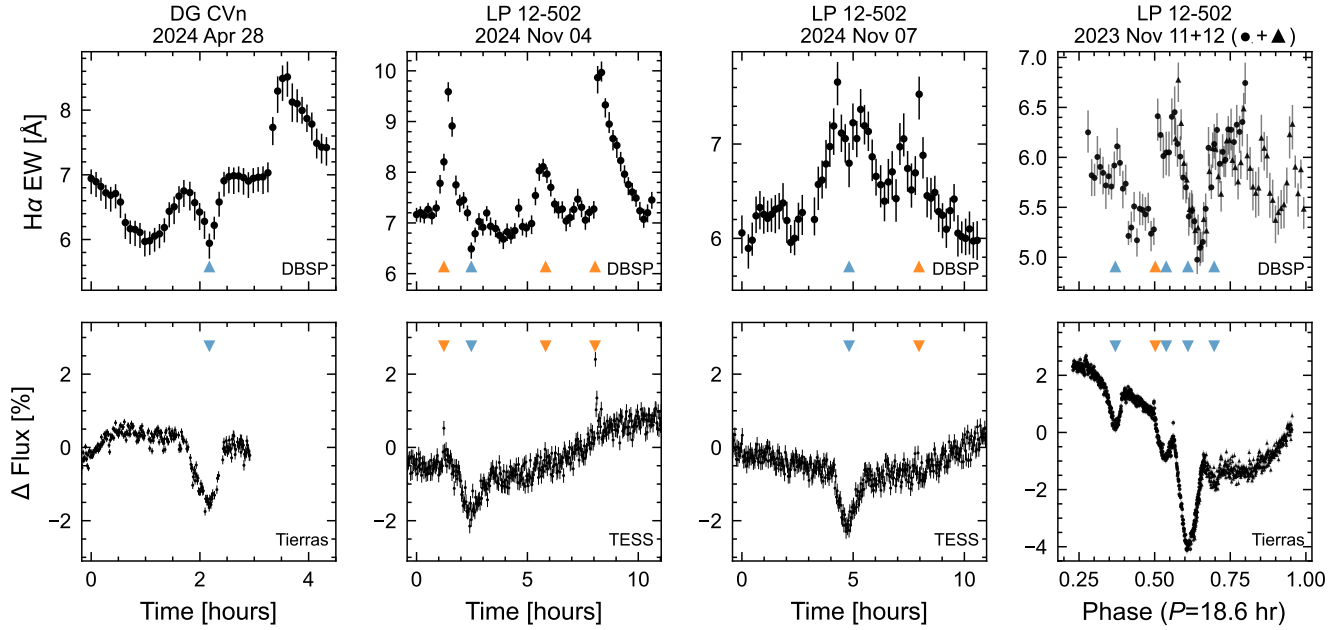


Figure 4. Balmer lines dim during dips and brighten during flares. Panels show Hale/DBSP (R≈2,200) Hα equivalent widths (EWs) vs. simultaneous Tierras and TESS photometry. Minute-timescale optical flares (orange arrows) are associated with hour-timescale Balmer EW flares, which dominate the Hα variability. The ∼hour timescale optical dips that define the CPV class (blue arrows) tend to be contemporaneous with Balmer EW dips, consistent with transiting clumps of dusty plasma (see discussion in Section 4.1). The rightmost column is shown as a function of phase since the data span two orbital cycles.

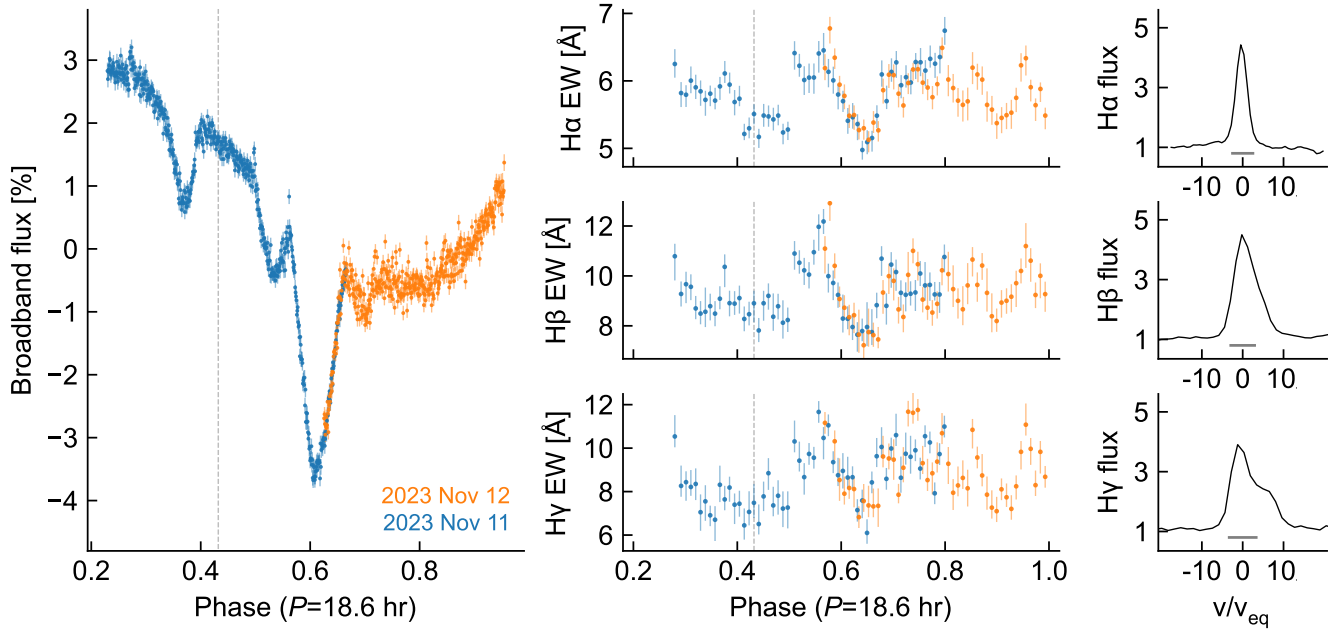


Figure 5. Hale/DBSP Hα, Hβ, and Hγ equivalent widths and line profiles of LP 12-502, 11–12 November 2023. This figure details Figure 4. The **online movie** shows the line profiles for each exposure, switching from Nov 11 to Nov 12 near ϕ≈0.8. The target, LP 12-502, has v_{eq}=24.2 km/s; gray horizontal lines denote a single velocity resolution element ($\delta v \approx c$/R). The Balmer line equivalent widths vary at the 5-50% level, coincident in time with 1-2% variability in the broadband Tierras filter. There is strong evidence for line shape differences between Hα, Hβ, and Hγ, and weak evidence for time variability in the velocity-resolved emission profile of Hγ. The jump at ϕ≈0.50 was likely flare-induced, based on similar timing coincidences in Figure 4.

per resolution element. The dispersed spectrum is recorded using a HAWAII-II HgCdTe detector.

We collected 13 spectra of LP 12-502 between 02:00 and 06:15 UT, each a 4×120 s ABBA sequence along the slit, allowing for the removal of the near-infrared sky background, terrestrial OH emission, and detector background through subtraction of adjacent nod positions. We observed A0V telluric standards at similar airmass for telluric correction and flux calibration. We performed flat-field corrections, wavelength calibration, spectral extraction, coaddition, telluric correction, and merging of the individual spectral orders using the IDL-based reduction package Spextool (M. C. Cushing et al. 2004).

We modeled each spectrum independently following F. Pérez Paolino et al. (2025a), representing the flux as

$$F_\lambda = \Big[(1-f_{\rm spot})\,F_\lambda(T_{\rm phot}) + f_{\rm spot}\,F_\lambda(T_{\rm spot}) + f_{\rm excess}\,B_\lambda(T_{\rm excess})\Big]\left(\frac{R_\star}{d}\right)^2, \quad (1)$$

where $F_\lambda(T)$ are BT-Settl synthetic spectra (F. Allard 2014) for the unspotted photosphere and the spots, $f_{\rm spot}$ is the spot filling factor, and B_λ is an optional blackbody with emitting area $f_{\rm excess}$ relative to the visible stellar disk. We sampled the posteriors with `dynesty` (J. S. Speagle 2020) under uniform priors, masked the telluric bands at 1.10–1.20, 1.325–1.45, and 1.80–2.05 μm, and added a 3% fractional model uncertainty in quadrature with the measurement errors. We repeated the fits with the spot term removed, and separately with the excess term removed, to assess whether the data required either.

2.7. *Magellan/FourStar*

We observed TIC 300651846 ($K = 10.8$) on 11 February 2025 (UT) using FourStar (S. E. Persson et al. 2013), a near-infrared wide-field imager on the Baade 6.5 m Magellan telescope. We used the NB2.09 narrowband filter ($\lambda = 2.082$–$2.118\,\mu$m), in the FullWell read mode, taking 20.38 s correlated double sampled exposures. The broader K-band filter would have saturated the detector given the target's brightness. Young M dwarfs present no prominent emission or absorption lines within the NB2.09 bandpass: the H_2 1–0 S(1) transition at 21,218 Å falls off-band. We therefore interpret variability detected in this filter as continuum variability.

Humidity kept us closed over the first quarter of the night. After opening, we observed for 5.3 hr beginning at $\mathrm{BJD_{TDB}} = 2460717.650$, with seeing averaging 0.″9. We reduced the images to light curves using `AstroImageJ` (K. A. Collins et al. 2017) and converted timestamps to $\mathrm{BJD_{TDB}}$ using the J. Eastman et al. (2010) barycentric correction code. Figure 6 shows the resulting light curve phased to $t_0 = \mathrm{BJD_{TDB}}$ 2460693.244 and $P = 0.34392$ d (8.25 hr), with the simultaneous TESS observation from Sector 88.

2.8. *MuSCAT1 and MuSCAT2*

We observed TIC 262400835 on 13 and 15 December 2020 UT using the MuSCAT2 multi-band imager (N. Narita et al.

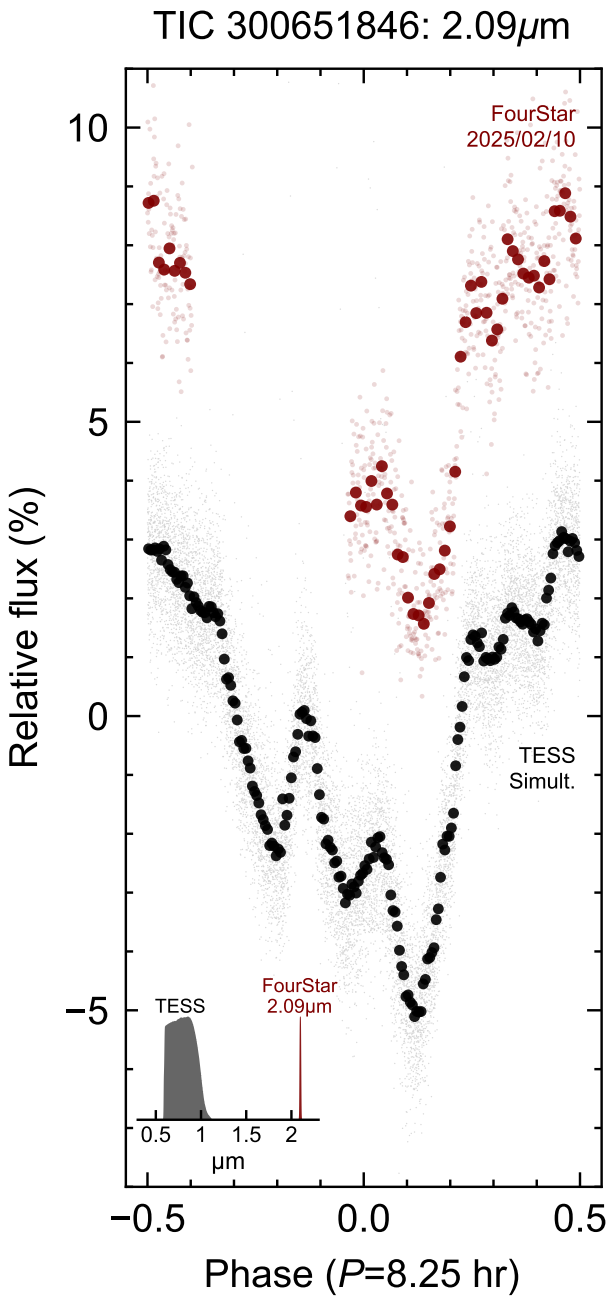


Figure 6. Magellan/FourStar 2.09 μm photometry of TIC 300651846 vs. simultaneous TESS (0.6–1 μm). The target has a period of 8.25 hours and was observable for two-thirds of a rotation cycle with FourStar on 11 February 2025. Simultaneous TESS data from Sector 88 are plotted underneath. The two bands (filter profiles in lower left) show dip depths in a ratio of 0.80±0.03 (NB2.09/TESS).

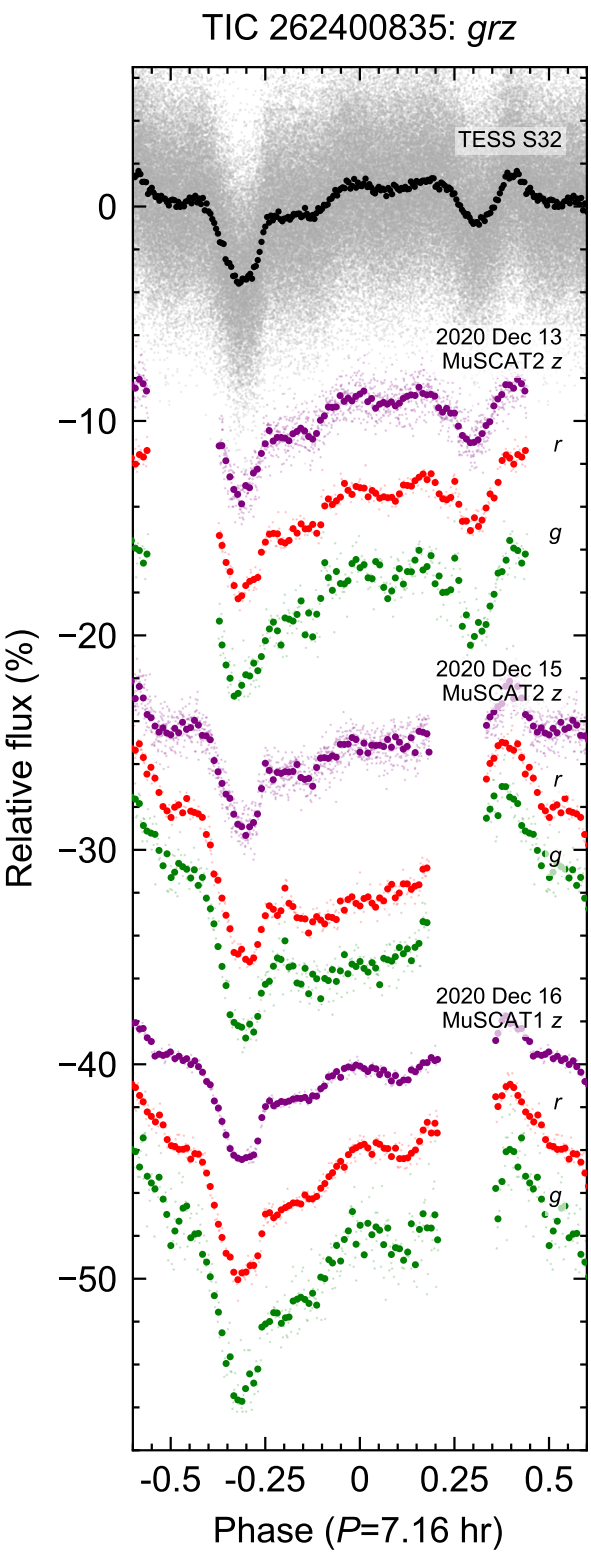


Figure 7. MuSCAT1 and MuSCAT2 $g'r'z_s$ photometry of TIC 262400835 vs. simultaneous TESS. The target has a period of 7.16 hours. The plotted TESS data, acquired at 20 second cadence, span 19 November 2020 to 17 December 2020.

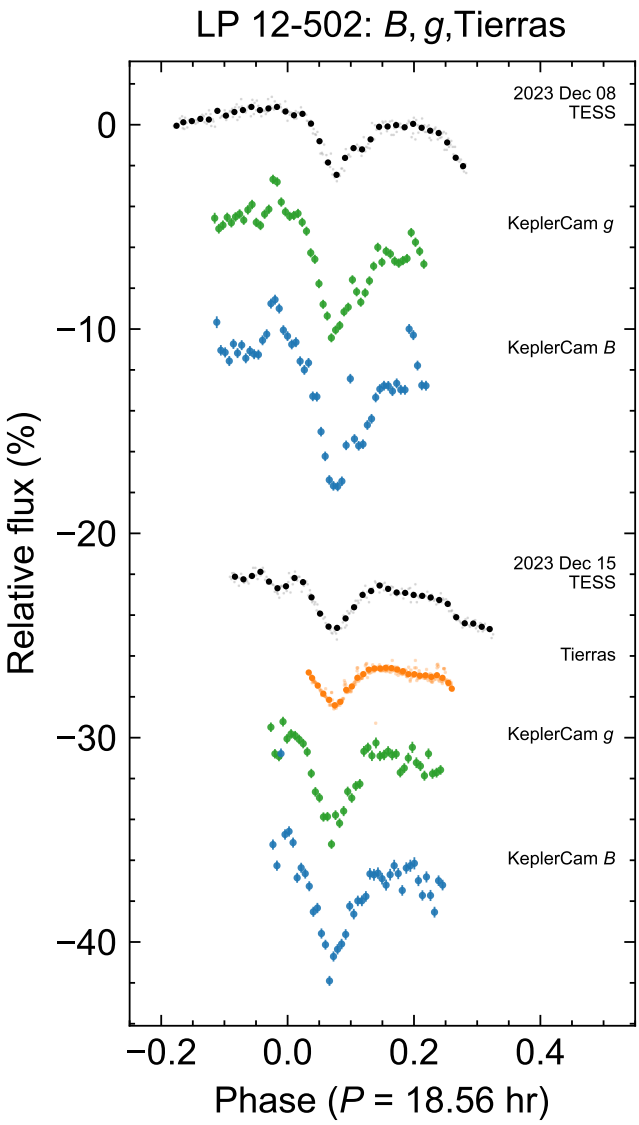


Figure 8. KeplerCam gB and Tierras photometry of LP 12-502 vs. simultaneous TESS. The target has a period of 18.56 hours. The plotted TESS data are excerpts of 120 second cadence data from Sector 73.

2019) mounted on the 1.5 m Telescopio Carlos Sánchez (TCS) at Teide Observatory, Spain. MuSCAT2 is equipped with four CCDs capable of obtaining simultaneous imaging in the g', r', i', and z_s bands with negligible readout overhead. Each CCD consists of 1024×1024 pixels, providing a field of view of $7.4 \times 7.4\,'^2$.

We conducted the MuSCAT2 observations with the telescope in nominal focus. On both nights, we set the exposure times to 60 s for the g', r', and i' bands, and 15 s for the z_s band. We processed the raw data using the MuSCAT2 pipeline (H. Parviainen et al. 2019), which performs standard dark and flat-field calibrations followed by aperture photometry. We also used this pipeline to model instrumental and

environmental systematics, specifically accounting for variations induced by airmass changes and pixel-positional shifts due to telescope guiding. We adopted identical reference stars across both MuSCAT2 nights, and adopted an aperture with r=16 px (7.″0) to ensure negligible contamination from TIC 262400836. This aperture choice localized the 7.16 hr "dip" signal to TIC 262400835.

We also observed TIC 262400835 on the night of 16 December 2020 UT using MuSCAT1 (N. Narita et al. 2015), a multi-band imager on the 1.88 m telescope of the National Astronomical Observatory of Japan (NAOJ) located in Okayama, Japan. MuSCAT1 is equipped with three CCDs, enabling simultaneous imaging in the g', r', and z_s bands with negligible readout overhead. Each CCD provides a field of view of 6.1×6.1 arcmin^2 over 1024×1024 pixels.

We used exposure times of 60 seconds in the g', r', and z_s bands. We dark-subtracted and flat-fielded the raw images, and performed aperture photometry following A. Fukui et al. (2011), adopting an aperture radii of 8, 10, and 14 px (2.″9, 3.″6, and 5.″1) to exclude the flux of the companion $\sim$10″ to the west. These observations were conducted simultaneously with TESS Sector 32, and the resulting light curves are shown in Figure 7 alongside the MuSCAT2 data.

2.9. *KeplerCam*

We observed LP 12-502 on 8 and 15 December 2023 UT using the KeplerCam CCD on the 1.2 m telescope at the Fred Lawrence Whipple Observatory (FLWO) at Mount Hopkins, Arizona. We took observations in alternating g' and B filters with 60 sec exposure in g' and 300 sec in B. KeplerCam is a 4096×4096 Fairchild CCD which has a field of view of $23.'1 \times 23.'1$ and an image scale of 0.672"/pix when binned by 2. We reduced the data using standard IDL routines, and performed aperture photometry using AstroImageJ (K. A. Collins et al. 2017). The resulting measurements are in Figure 8. We additionally observed on the night of 16 December 2023, when high winds and very poor seeing degraded the data quality. The KeplerCam B and g' light curves from that night weakly suggest the presence of a flux dip, and the simultaneous TESS light curve shows a $\approx$1% deep dip lasting an hour. Due to the poor quality of these data relative to the nights of 8 December and 15 December, we exclude them from further analysis.

2.10. *Tierras*

We observed LP 12-502 and DG CVn with the Tierras Observatory, a refurbished 1.3 m ultra-precise fully-automated photometer located at the F. L. Whipple Observatory atop Mount Hopkins, Arizona (J. García-Mejía et al. 2020). The facility carries out all observations through a single custom narrow bandpass filter (λ_C = 863.5 nm, FWHM = 40 nm) that minimizes precipitable water vapor (PWV) errors known to limit photometric precision from the ground (e.g. C. W. Stubbs et al. 2007; C. H. Blake et al. 2008; Z. K. Berta et al. 2012). Tierras is also equipped with a deep-depletion 4K × 4K e2v CCD (15 μm pixels) with a plate scale of 0.″43 pixel^{-1} operated in frame-transfer (shutter-less) mode, yielding a field of view (FOV) of $15' \times 29'$.

We observed LP 12-502 during the UT nights of 11 November, 12 November, and 15 December 2023, as well as DG CVn on the UT night of 28 April 2024. The telescope was in focus throughout. We exposed for 45 s on LP 12-502, collecting 556, 421, and 296 frames over 8.1, 6.1, and 4.2 hr, respectively, and for 30 s on DG CVn, collecting 731 frames over 7.9 hr. Our LP 12-502 observations sampled airmasses between 1.52 and 1.74, while our DG CVn observations sampled airmasses between 1.00 and 1.80. Skies were clear on all four nights, with relative humidities between 10% and 52%. The median image FWHM was 0.″9, 1.″7, 1.″3, and 1.″1, listed chronologically.

To extract light curves from the Tierras images, we first bias-corrected the images by subtracting bias levels derived from the overscan region of each detector half. We then computed an astrometric solution for each frame, and extracted photometry for stars with $G_{\rm RP} < 17$. We measured fluxes through a family of circular apertures spanning radii of 5 to 20 pixels, estimating the local sky in each case using annuli running from 35 to 55 pixels. Our per-frame error budget combined photon noise from the star and from the sky with dark current, read noise, and scintillation (G. Stefansson et al. 2017). We discarded frames in which the target flux encroached on the detector's non-linear regime, which is why only 609 of the 731 DG CVn frames were kept for the analysis. We retained all 1273 LP 12-502 frames.

Finally, we normalized our target fluxes by a weighted sum over the ensemble of comparison stars, with the weights set by the iterative scheme detailed in P. Tamburo et al. (2022). In this scheme, stars that contribute noise are progressively downweighted until the weights stop changing. We then freeze the converged weights and reuse them for every night of a given target. Holding the ensemble fixed in this way guarantees that a brightness change measured for a variable target between two nights cannot be an artifact of shifts in the comparison set. For each target we kept the aperture that minimized the measured scatter: 17 pixels (7.″3) for LP 12-502, with an ensemble of 403 comparison stars, and 10 pixels (4.″3) for the brighter DG CVn, with 86 comparison stars. Individual exposures for the two targets reach a photometric precision of 1.1 to 1.7 ppt.

2.11. *Dip Depth Measurement*

We measured the depth, mid-time, and duration of each dimming event in the ground-based photometry by treating every observed dip independently. We fitted the light curves using a two-stage least squares routine. We first manually provided a guess for which points were "in-dip", which neighboring points provided the "out-of-dip" baseline, and which points to exclude due to flaring. We constrained this initial guess to be identical across bandpasses. We then fit a smooth rotational baseline, $b(t)$, to only the out-of-dip points, and subsequently fit the residual, $f(t)-b(t)$, with a dip model. We explored various models for the rotational baseline (low-order polynomial; low-order Fourier series) and the

Table 3. Multicolor Photometric Observations

Star	Instrument	Night (UT)	Bandpasses	Dips Seen?	Chromatic?
TIC 262400835	MuSCAT2	2020 Dec 13	g', r', i', z_s, TESS	✓	?
TIC 262400835	MuSCAT2	2020 Dec 15	g', r', i', z_s, TESS	✓	✓
TIC 262400835	MuSCAT1	2020 Dec 16	g', r', z_s, TESS	✓	✓
LP 12-502	KeplerCam	2023 Dec 08	B, g'	✓	✓
LP 12-502	KeplerCam + Tierras	2023 Dec 15	B, g', Tierras	✓	✓
LP 12-502	KeplerCam	2023 Dec 16	B, g'	✓	?
TIC 300651846	FourStar + TESS	2025 Feb 11	NB2.09, TESS	✓	?

NOTE—"Dips Seen?" indicates whether sharp flux dips were observed; "Chromatic?" indicates whether the dip depths differed significantly between bandpasses. The FourStar entry is the central near-infrared constraint of this work.

dip model (hyperbolic secant per L. Yu et al. 2015; trapezoidal dip), always requiring the dip epoch and total duration to be identical over bandpasses, but letting the depth and shape parameters vary. We assessed the goodness of fit for each model using the reduced χ^2, as initially evaluated by setting the flux uncertainties to the point-to-point out-of-dip RMS (the $84^{\rm th}$-$16^{\rm th}$ percentile spread of the first differences $\Delta F_j = F_j - F_{j+1}$), and subsequently renormalized to force the best-fitting model to have $\chi^2_{\rm red} \equiv 1$. We then compared the model evidence using the Bayesian Information Criterion (BIC). We found across all the photometric datasets that baseline models of at most second order minimized the BIC, and that the trapezoidal dip model (parametrized by depth, duration, mid-time, and fractional ingress/egress duration) offered better $\chi^2_{\rm red}$ and ΔBIC than the hyperbolic secant (parametrized by depth, width, and mid-time). Table 4 lists the resulting depth measurements, based on the trapezoidal model and the BIC-preferred out-of-dip baseline models. Appendix B shows the resulting fits with each photometric dataset.

3. INTERPRETATION

3.1. *Bound circumstellar gas in emission*

The 18 January 2025 Magellan/MagE observations of TIC 300651846 (Figure 2) show sinusoidal time-varying Hα emission with semi-amplitude $K \approx 4\, v_{\rm eq}$. To our best knowledge, the only available explanation is emission from a cool ($\lesssim 10^4$ K) plasma clump trapped in corotation. This is the second CPV showing this phenomenology, after TIC 141146667 (L. G. Bouma & M. M. Jardine 2025). TIC 141146667 showed two clumps at opposite stellar longitudes, which is an azimuthal distribution predicted by models of rigidly rotating magnetospheres with tilted dipolar magnetic fields (R. H. D. Townsend 2008). Such models tend to produce "pinched" plasma tori[1]. For TIC 300651846, the clearest signal shows a single clump appearing at ϕ=0.1 near $\approx 4\, R_\star$ (beyond $R_{\rm c}$=2.49±0.15 $R_\star$), and rotating with the star for the next six hours. The clump emits with an EW ≈60% that of the star, and is internally unresolved at MagE's resolving power ($\approx 73\,{\rm km\,s^{-1}}$). A secondary feature in the 18 January 2025 night appears from ϕ=0.6-0.7. While this feature's interpretation is complicated by its brief lifetime, the star presents a triple-lined Hα profile in three separate exposures. This is consistent with the idea of opposing plasma clumps becoming visible at phase quadrature, when they are maximally visible.

We estimate the number density of the longer-lived plasma clump based on the emitting line's luminosity as follows. For the star, we observe a time-averaged Hα equivalent width of 6.0 ± 0.3 Å. Given the SED-derived $L_\star$=0.033±0.006 $L_\odot$ (L. G. Bouma et al. 2024), this implies that the stellar Hα line radiates $\approx 3.0 \times 10^{28}\,{\rm erg\,s^{-1}}$. From the same SED fit, the star has $R_\star$=0.61±0.04 $R_\odot$. Following Appendix C of L. G. Bouma & M. M. Jardine (2025) and approximating all the emission as coming from case B radiative recombination, a homogeneous sphere of radius $r \approx 0.1\, R_\star$ would yield $n_{\rm H} \approx 8 \times 10^{11}\,{\rm cm^{-3}}$, with a total clump mass $M_{\rm gas} \approx 5 \times 10^{17}$ g—all on the night of 18 January 2025.

The other nights in January 2025 showed much weaker circumstellar emission. Close inspection of the 19 January and 20 January MagE data (Figure 2) shows an emissive component appearing at similar orbital phases as on 18 January. The MIKE data from 2 January show a similar signal (Figure 3). However, this emission has an equivalent width ≈10% that of the star's average Hα profile, rather than ≈60%; on 21 January, no emission is detectable. Under radiative recombination, $n_{\rm H} \propto L_{{\rm H}\alpha}^{1/2}$, so this six-fold decrease in luminosity cor-

[1] See http://user.astro.wisc.edu/~townsend/static.php?ref=rrm-movies last accessed 26 Aug 2026.

Table 4. Dip depths by bandpass

Star	Date (UT)	Band	δ (%)	Ref.
TIC 262400835	2020 Dec 13	g'	3.36 ± 0.19	1
		r'	2.54 ± 0.10	1
		i'	2.74 ± 0.07	1
		z_s	2.69 ± 0.09	1
		TESS‡	0.96 ± 0.12	1
	2020 Dec 15	g'	4.67 ± 0.15	1
		r'	3.46 ± 0.10	1
		i'	3.64 ± 0.08	1
		z_s	2.99 ± 0.09	1
		TESS‡	1.95 ± 0.23	1
	2020 Dec 16	g'	4.30 ± 0.20	1
		r'	3.52 ± 0.08	1
		z_s	2.96 ± 0.06	1
		TESS‡	2.17 ± 0.38	1
LP 12-502	2023 Dec 08	B	5.57 ± 0.28	1
		g'	4.63 ± 0.18	1
		TESS	2.44 ± 0.08	1
	2023 Dec 15	B	4.48 ± 0.44	1
		g'	3.48 ± 0.24	1
		Tierras	2.75 ± 0.09	1
		TESS	2.18 ± 0.24	1
TIC 300651846	2025 Feb 11	TESS	4.07 ± 0.05	1
	(dip 1)	NB2.09	3.27 ± 0.10	1
	(+1.4 h)	TESS	0.62 ± 0.08	1
		NB2.09	1.06 ± 0.20	1
	(+2.1 h)	TESS	0.80 ± 0.05	1
		NB2.09	0.81 ± 0.10	1
TIC 435899024	2021 May 25–	B	12.47 ± 0.40	2†
	–2021 June 01	V	11.96 ± 0.41	2†
		R	10.79 ± 0.29	2†
		I	9.11 ± 0.21	2†
PTFO 8-8695	2014 Jan 19	I+z	2.18 ± 0.08	3†
		H	1.38 ± 0.05	3†
	2016 Feb 9	g'	2.57 ± 0.34	4†
		r'	1.69 ± 0.12	4†
		z_s	0.99 ± 0.10	4†
	2014 Feb 23	I_C	2.59 ± 0.10	5†
		K_s	1.07 ± 0.20	5†
	2014 Dec 27	I_C	1.25 ± 0.08	5†
		K_s	0.97 ± 0.11	5†
	2015 Jan 10	I_C	1.34 ± 0.13	5†
		K_s	1.24 ± 0.22	5†
	2015 Jan 23	I_C	1.56 ± 0.14	5†
		J	1.00 ± 0.22	5†
	2015 Feb 10	I_C	1.64 ± 0.15	5†
		H	0.87 ± 0.11	5†
	2015 Feb 14	I_C	2.16 ± 0.05	5†
		K_s	1.10 ± 0.10	5†
	2015 Feb 23	I_C	1.44 ± 0.17	5†
		K_s	1.12 ± 0.27	5†
	2016 Nov 25	I_C	1.01 ± 0.12	5†
		J	1.09 ± 0.11	5†
	2016 Dec 9	I_C	1.38 ± 0.15	5†
		J	0.90 ± 0.10	5†

NOTE—†Denotes literature photometry refitted in this work. ‡Listed for completeness but excluded from the extinction-law fits due to low S/N and an uncertain crowding correction (Section 3.3.1). *References*: (1) This work. (2) C. Koen (2023) SAAO 1m/CCD "J1252"; (3) L. Yu et al. (2015) Baade 6.5m/FourStar H and TRAPPIST I+z; (4) M. Onitsuka et al. (2017) NAOJ 1.88 m/MuSCAT simultaneous; (5) Y. Tanimoto et al. (2020) HHO 1.5 m/HONIR.

responds to a number density decrease of 2.5×. Despite this change in the circumstellar plasma number density, the simultaneous light curve morphology in Figure 2 appears unchanged.

The implication is that, at least over these five nights of observations, the light curve morphology was not directly tied to the circumstellar plasma number density. This is consistent with previous discussion (e.g. Section 5.6 of L. G. Bouma et al. 2024) arguing that there is no microphysical mechanism by which gas can provide the opacity needed to explain both the observed transit depths and their chromaticities. The spectra show that gas clumps are present—but dust remains the sole viable opacity source. The gas and dust broadly trace each other, in that Balmer dimming accompanies the broadband dips (Section 3.2). However, the large swings in gas emissivity without any change in dip morphology indicate that the dust column is not simply proportional to the gas density.

3.2. *Balmer dimming during transits*

With lower resolution spectra, it is no longer possible to separate circumstellar emission from the stellar chromosphere. Nonetheless, when a plasma clump transits, one would expect the plasma to scatter or absorb stellar Hα photons, yielding an Hα transit that mirrors the broadband flux. We refer to this combination of absorption and scattering (*i.e.*, extinction) as dimming. Figures 4 and 5 show low-resolution data that appear consistent with the hypothesis of greater Hα dimming during broadband transits: the local minima in flux are correlated with local minima in the Hα equivalent width time-series. Moreover, Figures 2 and 3 both show similar absorption in the Hα line cores of TIC 300651846 from ϕ=0.9–1.0, consistent with absorption or scattering from the same plasma clump that is then seen in emission at phase quadrature.

This type of low resolution data presents two difficulties. The first is flares: minute-timescale optical flares produce large-amplitude, hour-long Balmer flares. The orange markers in Figure 4 and in Appendix A.2 highlight such flare-induced variability, which complicates interpretation of Hα EW time-series data. The second challenge is connected to the *consistency* with which local minima in flux yield local minima in Hα EW. The broadband flux dips in the three leftmost columns of Figure 4 are all coincident in time with local minima in their corresponding Hα EW time-series. However, the nights of 11 and 12 November 2023 showed four local minima in flux. The largest of these local flux minima, which was observed on both nights at $\phi \approx 0.62$, corresponds to an Hα minimum that *lagged* the flux minimum on both nights by ≈50 minutes. The green "Hα core" lines in Figures 2 and 3 show an analogous offset with the opposite sign: for TIC 300651846, the deepest Hα minimum *led* the deepest broadband flux minimum by ≈25 minutes on all five nights in January 2025. While the origin of these time offsets is not clear, their repeatability suggests that they are a stable feature associated with the material that causes CPV dips.

Summarizing, the low-resolution observations show $\delta\approx$1-3% broad-band flux dips corresponding to ≈5-20% variations in Hα EW. If the occulting cloud were gas-free, and made only of dust, this would not happen: dust does not selectively absorb light at 6563 Å. These observations are therefore consistent with the idea that the transiting material must include hydrogen.

3.3. *A near-infrared break in the extinction law*

CPV dips tend to appear deeper in the blue than in the red (M. N. Günther et al. 2022; C. Koen 2023), which C. Koen (2023) attributed to extinction by sub-micron dust. Since methodological choices in detrending and dip fitting can affect the inferred chromaticity, we opted to measure the dip depths from both our new light curves and archival photometry using a single procedure.

3.3.1. *Extinction law methodology*

We collected literature photometry from observations of CPVs showing at least one dip in at least two simultaneous bandpasses. We required the dips to be visually distinguishable (based on $\mathrm{d}f/\mathrm{d}t$) from spot-induced modulation, which is also chromatic. We excluded non-simultaneous literature datasets even when they carried novel information (e.g. the Spitzer 4.5 μm light curve from D. R. Ciardi et al. 2015) due to the time-variable nature of CPV dip depths. Further, we excluded *(i)* the TESS-band measurements for TIC 262400835, due to their low S/N and uncertain crowding correction; and *(ii)* the simultaneous SPECULOOS-South and TESS photometry of TIC 20178925 from M. N. Günther et al. (2022), because its ground-based and TESS dips show different durations, which is inconsistent with the fitting assumptions we describe below.

The results are in Table 4. This data collection effort yielded 51 dip depth measurements spanning 0.44–2.16 μm over five stars and twenty distinct dip epochs. Half of these measurements are new from this work; the remainder are from TIC 435899024 (C. Koen 2023) and PTFO 8-8695 (M. Onitsuka et al. 2017; Y. Tanimoto et al. 2020).

We then fitted an extinction law to the dip depths by considering variants of a model in which

$$\delta \propto \lambda^{-\beta}, \tag{2}$$

where δ is the dip depth and β encodes how strongly the occultation deepens toward the blue. Positive β corresponds to deeper dips at shorter wavelengths, as expected for extinction by sub-micron dust; $\beta = 0$ is achromatic (gray), as expected for an optically thick occulter.

Working in logarithmic space, we defined $y_i = \log_{10}\delta_i$, yielding an uncertainty $\sigma_y \approx |\mathrm{d}y/\mathrm{d}\delta|\sigma_\delta = \sigma_\delta/(\delta \ln 10)$, for dip index i at an epoch $e(i)$. Here "$e(i)$" refers to the set of all multi-wavelength measurements of one dip at a given epoch, and "$s(i)$" refers to the set of all observed dips for any given star. We further found it convenient to define $\Lambda_i \equiv \log_{10}(\lambda_{\mathrm{eff},i}/\lambda_0)$, where λ_0 is an arbitrary reference wavelength, with a corresponding depth at that wavelength δ_0. We chose λ_0=0.788 μm, the effective wavelength of the TESS bandpass, and calculated the effective wavelengths of the various bandpasses using the SVO Filter Profile Service (C. Rodrigo & E. Solano 2020). We then considered four model variants:

$$\mu_i = c_{e(i)} - \beta\Lambda_i \quad \text{(single power law)}, \tag{3}$$
$$\mu_i = c_{e(i)} - \beta_{s(i)}\Lambda_i \quad \text{(per-star power law)}, \tag{4}$$
$$\mu_i = c_{e(i)} - \beta_{e(i)}\Lambda_i \quad \text{(per-epoch power law)}, \tag{5}$$
$$\mu_i = c_{e(i)} - \beta_1 w_i + (\beta_1 - \beta_2)h_i \quad \text{(broken power law)}, \tag{6}$$

where $c_{e(i)} \equiv \log_{10}\delta_{0,e(i)}$ is a fitted reference depth defined at the reference wavelength λ_0. For observations that have TESS measurements, $c_{e(i)}$ will be near the observed TESS depth; for those without, $c_{e(i)}$ will be an extrapolation based on whichever bands were observed. In the broken power-law model, $w_i \equiv \log_{10}(\lambda_{\mathrm{eff},i}/\lambda_b)$ is the log-wavelength offset from a free break wavelength λ_b, and $h_i \equiv \kappa\ln(1+e^{w_i/\kappa})$ is a smoothing function that ensures a continuous gradient in the transition from β_1 blueward of the break to β_2 redward of it. We take λ_b as the reference wavelength for the per-epoch amplitudes $c_{e(i)}$ rather than λ_0, and we set the smoothing scale to κ=0.01 dex, since our data do not constrain it. Finally, our priors on the fits were $\beta \sim \mathcal{N}(0,3)$, $c_{e(i)} \sim \mathcal{N}(\langle y_i\rangle, 1)$, and, for the broken power-law model, $\lambda_b/\mu\mathrm{m} \sim \mathcal{U}(0.7, 1.6)$. We assumed a Gaussian likelihood, $y_i \sim \mathcal{N}(\mu_i, \sigma_i)$, where we let $\sigma_i = (s_i^2 + \sigma_{\mathrm{int}}^2)^{1/2}$ optionally add an intrinsic scatter term to the per-dip uncertainties. We sampled using PyMC (O. Abril-Pla et al. 2023), and required $\hat{R} < 1.01$ with no divergences as an indicator of convergence.

3.3.2. *Extinction law results*

Figure 9 and Table 5 summarize the results. While Figure 9 shows the single power-law and broken power-law results in the $\delta/\delta_{0.788\,\mu\mathrm{m}}$–$\lambda_{\mathrm{eff}}$ plane, we emphasize that the fits are in the space of y_i vs. Λ_i. The normalization $\delta_{0.788\,\mu\mathrm{m}}$ is the model-derived reference depth at the TESS effective wavelength for each dip epoch in Equations 3–6. To provide a consistent normalization for visual comparison, we adopted the $\delta_{0.788\,\mu\mathrm{m},i}$ values from the "per-star" power law fits, a choice that favors neither of the two displayed models.

All four models agree that the dip depths are chromatic. The model comparison diagnostics also indicate that a single power-law extinction curve can be improved upon, either by introducing a population-wide flattening of the extinction toward the near infrared, or by invoking epoch-to-epoch or star-to-star variability in the power-law index. Key issues in the model comparison include whether the dip depth uncertainties s_i are reliable; how to weigh each model's parameter count; and the degree to which the data can select between the models.

If the dip depth uncertainties were unreliable, then we would want to incorporate an intrinsic scatter term σ_{int}, effectively down-weighting the most precise measurements and weakening our ability to make model-selection claims on a ΔBIC basis. We calibrated the reported depth uncertainties

Table 5. Extinction law fitting results ($\delta \propto \lambda^{-\beta}$), including goodness of fit and the mean $\pm$ std. dev. for β from the four models. The number of model free parameters k and degrees of freedom $\nu = N - k$ are given as no-$\sigma_{\rm int}$/with-$\sigma_{\rm int}$ pairs. Both χ^2_ν columns are evaluated against measurement error alone, s_i, not the value $\approx$1 obtained once the fitted $\sigma_{\rm int}$ is added in quadrature. ΔBIC is relative to Model 4 within each variant.

			χ^2_ν		ΔBIC		β	
Model	k	ν	no $\sigma_{\rm int}$	with $\sigma_{\rm int}$	no $\sigma_{\rm int}$	with $\sigma_{\rm int}$	no $\sigma_{\rm int}$	with $\sigma_{\rm int}$
Single	21/22	30/29	8.39	13.85	127.8	23.0	0.47 ± 0.02	0.46 ± 0.08
Broken [a]	23/24	28/27	7.25	12.31	76.6	16.3	—	—
β_1 (blueward)	—	—	—	—	—	—	0.68 ± 0.07	0.79 ± 0.16
β_2 (redward)	—	—	—	—	—	—	-0.10 ± 0.31	-0.09 ± 0.39
Per-star	25/26	26/25	5.43	10.58	32.9	7.7	—	—
TIC 262400835	—	—	—	—	—	—	0.51 ± 0.05	0.51 ± 0.18
LP 12-502	—	—	—	—	—	—	0.98 ± 0.07	0.97 ± 0.21
TIC 300651846	—	—	—	—	—	—	0.19 ± 0.04	-0.01 ± 0.14
TIC 435899024	—	—	—	—	—	—	0.55 ± 0.06	0.53 ± 0.32
PTFO 8-8695	—	—	—	—	—	—	0.59 ± 0.05	0.57 ± 0.10
Per-epoch	40/41	11/10	4.48	5.79	0.0[b]	0.0[b]	—	—
TIC 262400835, 2020 Dec 13	—	—	—	—	—	—	0.21 ± 0.10	0.28 ± 0.21
2020 Dec 15	—	—	—	—	—	—	0.60 ± 0.07	0.61 ± 0.19
2020 Dec 16	—	—	—	—	—	—	0.60 ± 0.09	0.61 ± 0.23
LP 12-502, 2023 Dec 08	—	—	—	—	—	—	1.32 ± 0.09	1.34 ± 0.21
2023 Dec 15	—	—	—	—	—	—	0.54 ± 0.11	0.65 ± 0.19
TIC 300651846, epoch 1	—	—	—	—	—	—	0.22 ± 0.04	0.22 ± 0.13
epoch 2	—	—	—	—	—	—	-0.54 ± 0.25	-0.54 ± 0.26
epoch 3	—	—	—	—	—	—	-0.01 ± 0.18	-0.01 ± 0.19
TIC 435899024, 2021 May 25–Jun 01	—	—	—	—	—	—	0.55 ± 0.06	0.53 ± 0.20
PTFO 8-8695, 2014 Jan 19	—	—	—	—	—	—	0.63 ± 0.12	0.63 ± 0.19
2014 Feb 23	—	—	—	—	—	—	0.87 ± 0.19	0.87 ± 0.22
2014 Dec 27	—	—	—	—	—	—	0.25 ± 0.14	0.25 ± 0.18
2015 Jan 10	—	—	—	—	—	—	0.08 ± 0.20	0.08 ± 0.23
2015 Jan 23	—	—	—	—	—	—	0.96 ± 0.50	0.94 ± 0.58
2015 Feb 10	—	—	—	—	—	—	0.85 ± 0.21	0.85 ± 0.26
2015 Feb 14	—	—	—	—	—	—	0.67 ± 0.10	0.67 ± 0.15
2015 Feb 23	—	—	—	—	—	—	0.26 ± 0.26	0.25 ± 0.29
2016 Feb 9	—	—	—	—	—	—	1.50 ± 0.25	1.50 ± 0.32
2016 Nov 25	—	—	—	—	—	—	-0.16 ± 0.34	-0.16 ± 0.43
2016 Dec 9	—	—	—	—	—	—	0.93 ± 0.34	0.93 ± 0.43

[a] Break wavelength $\lambda_b = 1.22 \pm 0.31\,\mu$m (no $\sigma_{\rm int}$), $1.18 \pm 0.27\,\mu$m (with $\sigma_{\rm int}$).

[b] All four models suffer pathologies connected to the need to fit the reference depth terms "$\delta_{0,i}$". However, this problem is most severe for the per-epoch model, since 13 of 20 epochs have two bands and are therefore fit exactly (zero residual χ^2 regardless of the true β). This model's minimization of ΔBIC is an artifact of the incidental-parameters problem, not evidence of real epoch-to-epoch variation. See Section 3.3.2.

as described in Section 2.11, using the empirical out-of-dip scatter in each time-series dataset. The data and fitted model uncertainties (see Appendix B) suggest that this procedure is indeed yielding uncertainties that agree with the scatter in the data. As one example, our refit to the archival literature photometry from M. Onitsuka et al. (2017) yielded statistical uncertainties 10-17$\times$ larger than the published values, mirroring the observed scatter. Our dip depth uncertainties therefore seem plausible.

Nonetheless, there is an issue with treating our dip depth uncertainties at face value: both the universal power-law and the broken power-law models provide poor fits to the data (χ^2_ν=8.39 and 7.25, respectively). While inspection of χ^2_ν and ΔBIC in Table 5 would suggest that letting β vary on a per-epoch basis is preferable, that alternative also comes with drawbacks. In particular, the "β per-epoch" model is pathological for dips observed with only two simultaneous bandpasses, comprising over half the dataset: this model precludes χ^2 or BIC from accruing any residual informa-

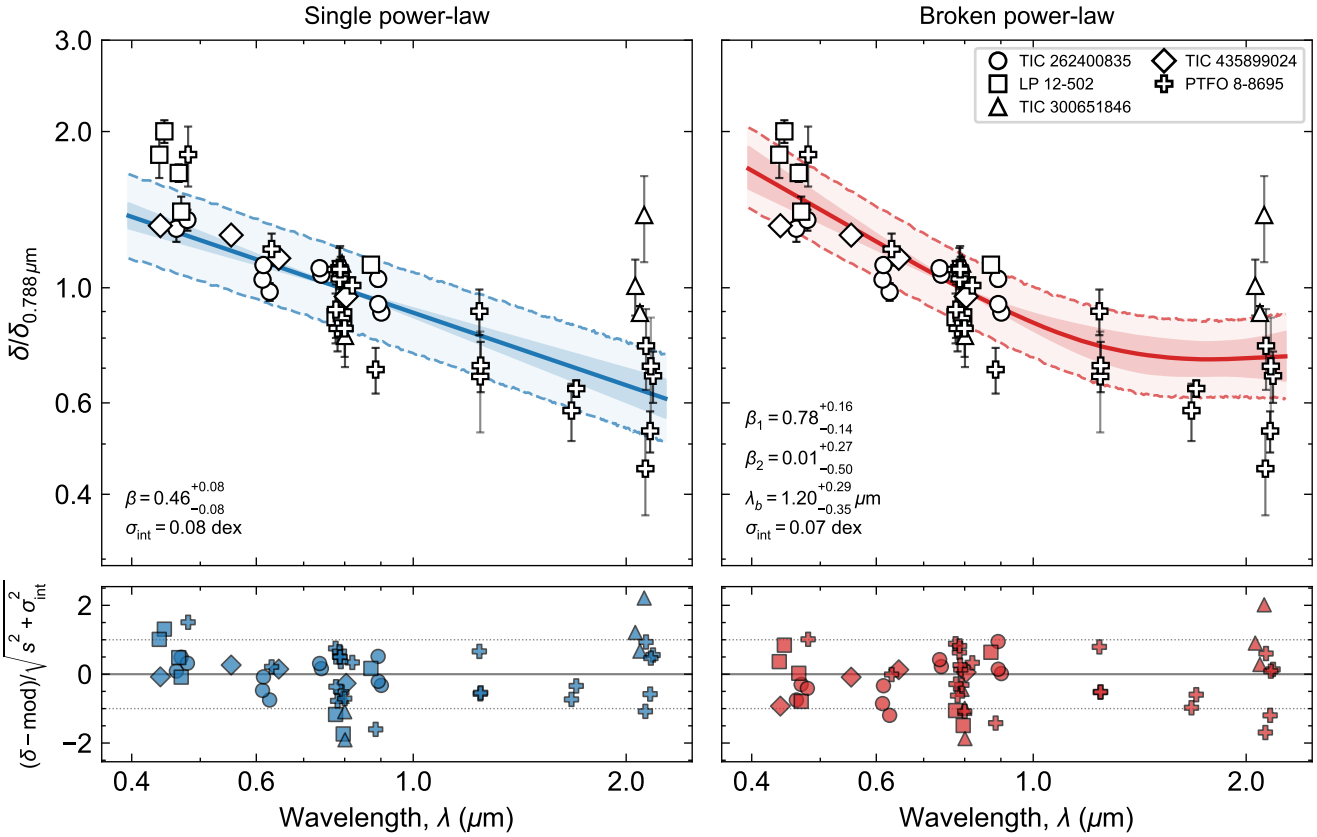


Figure 9. CPV dips are chromatic, with a near-infrared break. Markers show the measured dip depth in each bandpass from Table 4, with small horizontal jitter added for visual clarity. The normalization factor $\delta_{0.788\,\mu m}$, the depth at TESS-band (λ=0.788 μm), is a model-derived quantity per combination of star and observing epoch (see Section 3.3). The solid line and inner band are the posterior-median and the 68% confidence interval of the mean $\delta(\lambda)$ relation; the outer band is the 68% predictive interval after additionally accounting for intrinsic variability (σ_{int}). Including σ_{int}, there is a positive preference (ΔBIC=6.7) in favor of the broken power-law. Excluding σ_{int}, there is a strong preference for the broken power-law (ΔBIC=51.2). *References*: TIC 262400835, LP 12-502, and TIC 300651846: this work. TIC 435899024: C. Koen (2023). PTFO 8-8695: L. Yu et al. (2015), M. Onitsuka et al. (2017), Y. Tanimoto et al. (2020).

tion because it adds free parameters at the same rate as new data points. Ultimately, we found that with this dataset we were unable to produce a reliable ranking of this "per-epoch" model compared against the other models. To some degree, all four models suffer this "incidental parameters problem" (T. Lancaster 2000), in that every new observation epoch requires fitting and marginalizing over the depth at the reference wavelength; the problem is just most severe for the per-epoch model.

In brief, two divergent options can improve on the single power law: one can introduce a population-wide break that flattens the extinction toward the near-infrared, or one can invoke epoch-to-epoch or star-to-star variability. Assuming that the extinction law is universal, the preferred model is a broken power-law with $\beta_1 = 0.79 \pm 0.16$ blueward of the break at $\lambda_b = 1.18 \pm 0.27\,\mu$m, and $\beta_2 = -0.09 \pm 0.39$ redward of it. Intuitively, this model seems to provide the most plausible explanation for the narrowband FourStar 2.09 μm observations shown in Figure 6, which yielded dip depths $0.80\pm0.03\times$ the TESS depth, compared to a predicted ratio from the single-β fit of 0.64 ± 0.05 with σ_{int} (2.9σ), or 0.63 ± 0.01 without σ_{int} (5.9σ). With that said, we cannot formally exclude the possibility that TIC 300651846 is a low-β star with no extinction-law break.

3.4. *No evidence for extreme starspot filling factors*

The LP 12-502 TripleSpec spectra show no evidence for a spectroscopically distinct cool spot component. Fits with and without the T_{spot} term in Equation 1 yield indistinguishable χ^2_ν, and in the two-temperature fits the posteriors collapse to $T_{spot} \approx T_{phot}$. Fully-convective $\approx$3100 K stars at this age have spot-to-photosphere temperature ratios of $\approx$0.9 and filling factors f_{spot} of 0.1–0.6 (F. Pérez Paolino et al. 2025a). Spots with lower contrast are harder to detect through this method. Since these models recover filling factors down to $f_{spot}\approx$0.2 (F. Pérez Paolino et al. 2025a), we adopt $f_{spot} \lesssim 0.2$ as the limit for spots with temperature contrasts large enough to be detected on LP 12-502.

The TripleSpec data do, however, require an excess component. Relative to a photosphere-only model, adding it lowers χ^2_ν from 1.38 to 0.90, and is favored at ΔBIC=1372 and $\Delta \ln Z$=686. The component has $T_{excess} = 3827 \pm 271$ K and

$f_{\rm excess} = 0.17 \pm 0.08$ (median and scatter across spectra), and has an emitting area equivalent to a $0.13\,R_\odot$ sphere. With it, the photosphere has $T_{\rm phot} = 3074 \pm 50$ K, consistent with Table 1, and $R_\star \approx 0.32\,R_\odot$, slightly below the SED-based value; without it, $T_{\rm phot}$ (based on TripleSpec alone) rises to $\approx$3200 K and the fit degrades. The component is hotter than the photosphere and survives when a cool-spot term is allowed, so it is not starspot contamination; it seems too hot to be thermal dust emission; its persistence over four hours argues against a flare; and its emitting area is far too small for a stellar companion of this temperature. None of the $\approx$100 stars fit with the same models (F. Pérez Paolino et al. 2025a,b,c) show a comparable excess. The excess may indicate warm, optically thick material above the photosphere, though we cannot exclude systematic errors in the synthetic atmospheres. Distinguishing these possibilities, and establishing whether the excess is generic to CPVs, would require near-infrared spectroscopy of additional CPVs with simultaneous coverage blueward of 1 μm, where the component's fractional contribution to the flux is largest.

4. DISCUSSION

4.1. *An opaque core and an optically thin halo of dusty plasma*

It is tempting to read Table 5 and Figure 9 as measurements of the dust opacity law, $\kappa(\lambda)$, whose shape is connected to the size distribution and composition of the dust grains. While $\delta(\lambda)$ and $\kappa(\lambda)$ are related, there is a nuance: the mapping is only direct when the occulting material is optically thin. Nothing about the data guarantees that we are in this regime. Consider an absorbing clump of material occulting a uniform stellar disk of area $A_\star$. The dip depth δ is

$$\delta(\lambda) = \frac{1}{A_\star} \int \left[1 - e^{-\kappa\Sigma}\right] {\rm d}A, \tag{7}$$

where Σ is the surface density of the dust, κ is the opacity (and only wavelength-dependent term), and the integral is over the dust clump's area. Approximating $\kappa(\lambda) \propto \lambda^{-\beta}$ in the optical and near-infrared, one can show that the *observed* slope $\beta_{\rm obs}$ from $\delta(\lambda)$ in Figure 9 is related to the *underlying* dust opacity law's β through

$$\beta_{\rm obs} \equiv -\frac{{\rm d}\ln\delta}{{\rm d}\ln\lambda} = \beta \frac{{\rm d}\ln\delta}{{\rm d}\ln\kappa}. \tag{8}$$

The derivative on the right-hand side is a "dilution factor" which lies between 0 and 1 for any surface density distribution. The observed dip-depth $\beta_{\rm obs}$ of Table 5 and Section 3.3 therefore provides only a lower bound on the dust opacity's β, with the two related by Equation 8.

Figure 10 quantifies this damping for representative surface-density profiles and mean optical depths. Note that while the "uniform sphere" profile has $\Sigma(r) \propto (R^2 - r^2)^{1/2}$, this is not visible in the top panels due to the color stretch. The Gaussian has width w such that $\Sigma(r) \propto \exp(-r^2/(2w^2))$. The middle panel shows that for the smooth profiles, by a mean optical depth $\bar{\tau} \approx 1$ the observed slope falls to 40–60%

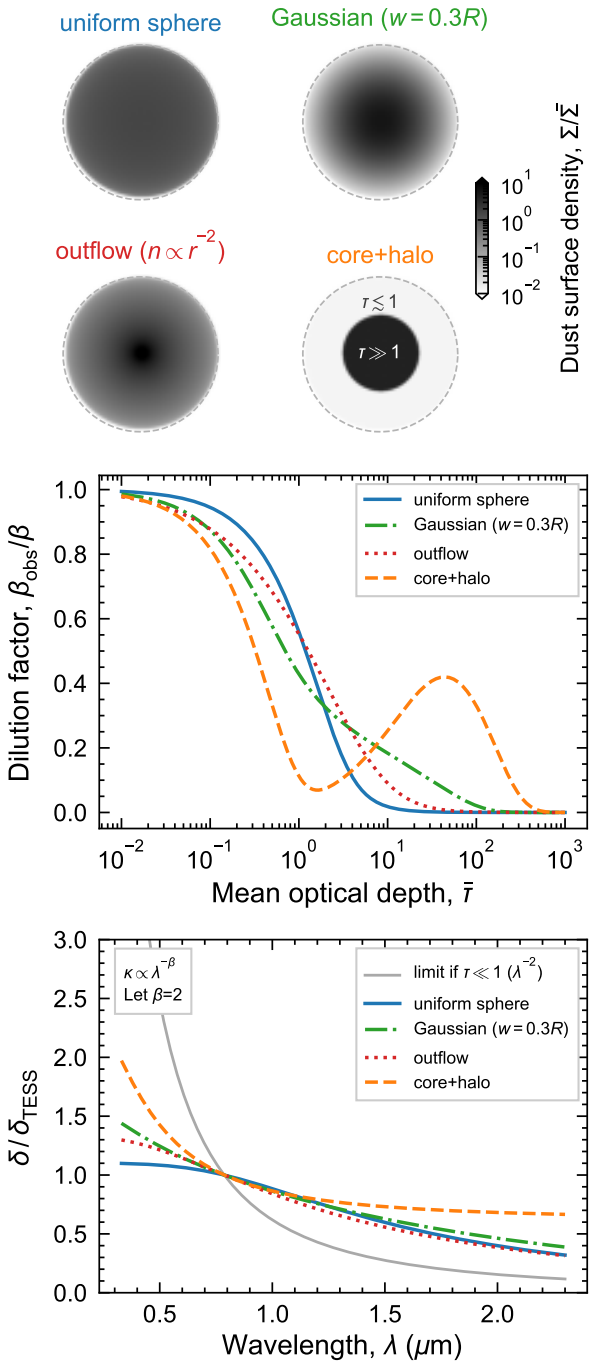


Figure 10. Partial opacity dilutes chromaticity. *Top*: surface density maps for four example clump profiles. The "uniform sphere" has constant number density; the "core+halo" (Equation 9) has $A_{\rm h}/A_{\rm c}$=3. *Middle*: the factor by which an intrinsic dust opacity $\kappa \propto \lambda^{-\beta}$ is diluted into an observed dip chromaticity, $\delta \propto \lambda^{-\beta_{\rm obs}}$ (Equation 8). The mean optical depth, $\bar{\tau}$, is averaged over the clump's projected disk. At fixed R and κ, profiles with equal $\bar{\tau}$ contain equal dust mass. *Bottom*: example spectra for β=2; the smooth profiles set $\bar{\tau}_{\rm TESS}$=3, and the core+halo profile sets $\tau_{\rm h}$=0.1 at 1.2 μm. Only the bimodal core+halo profile matches Figure 9.

of the intrinsic slope, and by $\bar{\tau}\approx3$ to 20–30%. The lower panel gives example vertical slices from the middle, setting $\bar{\tau}=3$ for the smooth profiles and exploring the impact of dilution on a "steep" $\beta=2$ intrinsic grain opacity. All models flatten the extinction curve. However, even after accounting for this dilution, comparison with Figure 9 shows that the smooth profiles provide a poor fit to the data: their blue dips are not deep enough, and their K-band dips are too shallow.

If dust provides the opacity, the simplest physical interpretation of a broken power-law in $\delta(\lambda)$ is therefore to invoke a *bimodal* surface density distribution: an opaque core of projected area A_c ($\tau \gg 1$ out through K band) surrounded by an optically thin halo of area A_h and depth $\tau_h(\lambda)$. Such a distribution yields

$$\delta(\lambda) \approx \frac{A_c}{A_\star} + \frac{A_h}{A_\star}\left[1 - e^{-\tau_h(\lambda)}\right], \tag{9}$$

such that the infrared depths sit on the core's gray floor while the optical depths are dominated by the halo's chromatic term, with the break falling near where the two terms are comparable. The "core+halo" curve in the lower panel of Figure 10 shows an example with $A_h/A_c=3$ and $\tau_h(1.2\,\mu m)=0.1$: the same $\beta=2$ opacity law that the smooth profiles flatten into a gray curve instead yields a broken power law resembling the data. A brute-force exploration of the parameter space indicates an allowed parameter range of $1 \lesssim A_h/A_c \lesssim 30$, $10^{-4} \lesssim \tau_h \lesssim 0.3$ at $1.2\,\mu m$, and $1 < \beta < 4$, where the upper bound on β is the Rayleigh-scattering limit. This range of β excludes the possibility of a population dominated by mm to cm-sized solids, but is otherwise consistent with a wide range of dust compositions and size distributions.

4.2. *Where do the dust and plasma come from?*

Where does the occulting material responsible for CPV dips come from? Figure 11 sketches three possible scenarios: *(i)* dust-free plasma clumps combined with starspots, *(ii)* dust that condenses out of the magnetically confined stellar plasma, and *(iii)* dust captured from an extrinsic reservoir, such as a debris disk or undetected planetesimals. All three scenarios require material in corotation beyond R_c, held in the local potential minima along closed field lines (R. H. D. Townsend & S. P. Owocki 2005), with $R_m > R_c$ so that magnetic tension enforces corotation at the clump locations. The sinusoidal Balmer emission supports this: for TIC 300651846 and TIC 141146667, the clumps emit at $\approx$1.1–2.1 R_c. Starspots are present in every scenario, since CPVs are young, magnetically active, and show smooth rotational modulation over which the dip signals are superposed. While dark facular networks could also be present (A. I. Shapiro et al. 2026), they are relevant only in that they could influence this smooth variability.

In scenario *(i)*, the plasma clumps condense out from the star (e.g. S. Daley-Yates & M. M. Jardine 2024), and a very specific starspot geometry is required. The chromaticity must be provided by the plasma, the spots, or a combination thereof. The plasma cannot do it alone: neither H^-

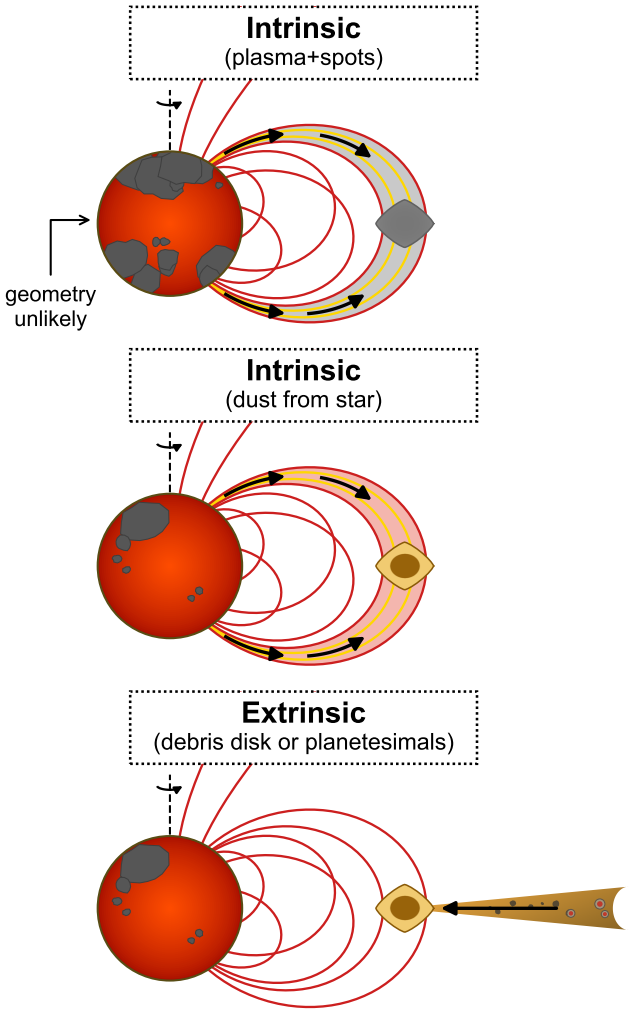


Figure 11. Dust is required in the viable origin scenarios. *Top*: corotating plasma clumps plus high starspot filling factor and no dust. *Center*: Dust condensing from magnetically confined stellar plasma. *Bottom*: Dust captured from an extrinsic reservoir (debris disk or undetected planetesimals). The dust-free scenario is disfavored; the lower two remain possible, and differ in where the dust originates. At least some starspots are present in all scenarios. The stars must have "centrifugal" magnetospheres with magnetospheric radii R_m exceeding their corotation radii R_c; the clumps remain long-lived only between the two extremes.

opacity nor hydrogen bound-free opacity provides the correct functional form. The most likely plasma opacity source would instead be Thomson scattering from the free electrons, combined with the "transit light source effect" (B. V. Rackham et al. 2018), in which unocculted spots outside the transit chord provide the apparent chromaticity. Exploring such models, however, shows that the required spot-covering fractions are 70–90% of the stellar area, while the clump sizes required by CPV dip depths ($R/R_\star \gtrsim 0.1$) leave little room for clear transit chords. The geometric probability of clear chords across the entire ensemble of observed chromatic

CPV dips is low enough to rule this scenario out. Separate arguments against it include the lack of light curve response to changes in Hα emission (Section 3.1), and the evidence from TripleSpec for starspot covering fractions $\lesssim$20% in LP 12-502 (Section 3.4).

While we disfavor scenario *(i)*, the data remain compatible with scenarios *(ii)* and *(iii)*. In either, the dust must become collisionally charged (H. Sanderson et al. 2023) to remain magnetically confined (S. Daley-Yates et al. 2026). The two scenarios may differ in their expected extent of mixing between the gas and dust, and in the composition of the grains. Either origin channel would yield new information: in scenario *(ii)* we would learn that young M dwarfs condense out dust, similar perhaps to how this process occurs in the atmospheres of AGB stars (S. Höfner & H. Olofsson 2018); in scenario *(iii)* we would gain a new way of identifying and studying the detritus of planet formation.

4.3. *Continued relevance of PTFO 8-8695*

The saga of PTFO 8-8695 (CVSO 30), whose data appear in Figure 9, provides unique connections to the current CPV literature. To recap: J. C. van Eyken et al. (2012) identified a hot Jupiter candidate in Orion; J. W. Barnes et al. (2013) and S. Kamiaka et al. (2015) subsequently argued that its orbit was nodally precessing, and C. M. Johns-Krull et al. (2016) found that the source showed sinusoidal-in-time Hα emission at circumstellar velocities. Photometric follow-up found that the planet candidate's orbital period stayed fixed, but that the signal jumped around in orbital phase and split into multiple dips per cycle (D. R. Ciardi et al. 2015; L. Yu et al. 2015; S. Raetz et al. 2016; M. Onitsuka et al. 2017; Y. Tanimoto et al. 2020). L. G. Bouma et al. (2020) used TESS and Gaia to argue that the system shows multiple indicators for being an intermediate-separation (multiple AU) binary, one component of which shows an ordinary rotation signal, and the other component being a complex periodic variable. While this interpretation may not be uniformly accepted (T. A. Grosson & C. M. Johns-Krull 2021), further photometric follow-up by R. Bischoff et al. (2022) has been broadly consistent.

The characterization of PTFO 8-8695 provided photometric and spectroscopic details that have been remarkably prescient for understanding CPVs. Spectroscopically, C. M. Johns-Krull et al. (2016) showed evidence for Balmer variability similar to that seen for TIC 141146667 and TIC 300651846, and consistent with the idea of bound circumstellar plasma being present. Photometrically, Y. Tanimoto et al. (2020) monitored the source for five years in optical and infrared bands, and observed a break in the chromaticity of this system's dip depths going out to *K*-band similar to that reported in this work. One of their proposed resolutions was the scenario of an opaque dust core with an optically thin dust halo; this is the same scenario that we favor in Section 4.1. The Spitzer 4.5 μm light curve observed by D. R. Ciardi et al. (2015) showing a 0.5% deep dip in 2012 remains a unique dataset, as it provides the only information we have about the chromaticities of any CPV dips redward of *K*-band.

4.4. *The chromaticity differs from analog magnetic rotators*

Massive magnetic B stars are the only class of star known to exhibit similar light curves to CPVs. The prototype, σ Ori E, has a long history (e.g. J. D. Landstreet & E. F. Borra 1978; R. H. D. Townsend et al. 2005; M. E. Oksala et al. 2015), but the summary is that it hosts a centrifugally supported magnetosphere in which a long-lived "pinched" plasma torus (R. H. D. Townsend 2008) produces photometric eclipses and sinusoidal-in-time spectroscopic emission. L. G. Bouma et al. (2024) argued that the additional step to produce "complex" light curves comes from magnetic topology, on the basis that the few σ Ori E analogs showing complex light curves also show (rare) evidence from Zeeman Doppler imaging for non-dipolar magnetic fields. The analogy extends to the radio: σ Ori E shows rotationally modulated gyrosynchrotron emission from its corotating magnetosphere (P. Leto et al. 2012), and polarized, variable radio emission has now been reported for two CPVs, DG CVn (S. Kaur et al. 2025) and TIC 146539195 (S. Kaur et al. 2024; S. Curiel et al. 2026).

While physical analogies can be drawn, the chromaticities of the σ Ori E analog eclipses are different from what is seen for CPVs. J. E. Hesser et al. (1977) and M. Oksala & R. H. D. Townsend (2007) provided photometry that showed near-gray chromaticity in *vby*, with a slight depth decrease in *v* relative to *y*, and a major depth increase in *u*. This behavior mirrors the hydrogen bound-free opacity curve as a function of wavelength (D. F. Gray 2008): the Balmer break at 3646 Å captures the *u* band, while *vby* would have a slight decrease before the break. This is consistent with the opacity in the σ Ori E analogs coming purely from trapped hydrogen. While *u*-band photometry of CPVs does not exist, their behavior in the *griz* optical bands is inconsistent with the predictions of the bound-free opacity.

5. CONCLUSIONS

CPVs show smooth rotational modulation together with sharp dips from circumstellar material. The material must be circumstellar given how quickly the flux changes, and magnetically bound, given its synchronicity with the stellar rotation period and its response to flares (J. Stauffer et al. 2017; L. G. Bouma et al. 2024). A priori, the material could be gas (*i.e.*, plasma), dust, or both. However, in Section 4.2 we rule out the "gas-only" scenario: neither H^- nor hydrogen bound-free opacity yields the observed wavelength dependence, and the remaining option, Thomson scattering with chromaticity supplied by unocculted starspots, requires spot covering fractions that are excluded. While gas is conspicuously present, dust must provide the opacity.

Gas is conspicuous for two reasons: sinusoidal-in-time Balmer emission out of transit, and Balmer dimming during transit. The emission yields a measurement of the hydrogen number density of the bound material at a few stellar radii. After TIC 141146667 ($n_{\rm H}\approx1\times10^{11}$ cm^{-3}; L. G. Bouma & M. M. Jardine 2025), TIC 300651846 shows emitting plasma at $n_{\rm H}\approx3$–8×10^{11} cm^{-3} (Section 3.1), and LP 12-502, observed at comparable sensitivity, gives $n_{\rm H}\lesssim5\times10^{10}$ cm^{-3}

(Appendix A.1). The range for TIC 300651846 reflects a six-fold change in clump Hα luminosity, during which the simultaneous TESS light curve did not discernibly change. The dip morphology is therefore decoupled from the plasma density, pointing to dust as the main opacity source. Balmer dimming during broadband dips, meanwhile, appears in every CPV observed spectroscopically during a dip, with repeatable timing offsets of tens of minutes in two stars (Section 3.2). The gas and dust are spatially associated, but the dust column does not track the gas density.

Broadband chromaticity currently provides the best information about the dust. We combined our new observations with the literature and fitted four variants of a $\delta \propto \lambda^{-\beta}$ extinction curve. While all four models agree that the dips are chromatic, a simple power law provides a poor fit ($\chi^2_\nu \gg 1$). Two modifications can improve it: the extinction law can flatten in the near-infrared, or there can be star-to-star or epoch-to-epoch variability in β. The data currently cannot separate the two due to the limited resolution of the multi-band photometry: future *spectrophotometry* would solve this issue. If at present we assume a universal law, then a broken power-law is preferred, with $\beta_1 = 0.79 \pm 0.16$ blueward of $\lambda_b = 1.18 \pm 0.27\,\mu$m and $\beta_2 = -0.09 \pm 0.39$ redward.

The break, rather than the slopes, carries most of the physical information (Sections 4.1 and 4.2). The *observed* optical chromaticity β_1 provides a lower bound on the *intrinsic* spectral index of the dust opacity: for $\kappa \propto \lambda^{-\beta}$ we infer $1 < \beta < 4$. This excludes a dust population dominated by mm to cm-sized solids, but is otherwise consistent with a wide range of dust compositions and size distributions. The break itself can be produced by a bimodal dust surface density distribution: an opaque core surrounded by an optically thin halo. We find that such a model can reproduce the data for halo to core area ratios $1 \lesssim A_\mathrm{h}/A_\mathrm{c} \lesssim 30$ and halo optical depths $10^{-4} \lesssim \tau_\mathrm{h} \lesssim 0.3$ at $1.2\,\mu$m. In this picture, a minority of the projected area is opaque through *K* band and the remainder is nearly transparent.

Gas is conspicuous: it emits in corotation, dims during transit, and changes in density without altering the dips. Dust is cryptic: it must supply the opacity, yet whether it condenses from the star or is captured from planetesimal debris (Section 4.2) is unknown. The core+halo picture of a dusty plasma makes three key predictions. First, dips should be gray redward of λ_b out to whatever wavelength the core becomes optically thin. Second, dip depths should vary smoothly across the Balmer break, unlike the *u*-band jump of the hydrogen-dominated σ Ori E analogs (Section 4.4). And third, if the grains contain silicates, the 10 μm Si-O stretching feature should appear as excess depth above the gray floor, with a profile that could distinguish wind condensates from collisional debris.

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Contributions: LGB led the design, execution, and interpretation of the project. FPP performed the TSPEC analysis. Data were acquired and reduced by JHL, KI, NN, and AF (MuSCAT1); EP, FM (MuSCAT2); JGM, PT, DC (Tierras); AB (KeplerCam); FPP (TSPEC); and LGB (MIKE,

MagE, FourStar, HIRES, DBSP, TSPEC). Observations by JJS (FourStar) and ASC (HIRES), and reduction assistance by HI (HIRES) enabled the work. Finally, FPP, LAH, JJS, MMJ, and DRC contributed to the interpretation of the results, and all authors reviewed the manuscript.

Facilities: Magellan:Baade, Magellan:Clay, Keck:I, Hale, NAOJ:1.88m, Sanchez, FLWO:Tierras, FLWO:1.2m, Gaia, TESS, 2MASS, SDSS.

Software: astroariadne (J. I. Vines & J. S. Jenkins 2022), astropy (Astropy Collaboration et al. 2013, 2018, 2022), carpy (D. D. Kelson et al. 2000; D. Kelson 2003), Claude (Anthropic 2026), DBSP_DRP (M. S. Mandigo-Stoba et al. 2022a,b), dustmaps (G. Green 2018), dynesty (J. S. Speagle 2020), matplotlib (J. D. Hunter et al. 2007), numpy (S. Van Der Walt et al. 2011), PyMC (O. Abril-Pla et al. 2023), PypeIt (J. Prochaska et al. 2020), scipy (P. Virtanen et al. 2020).

APPENDIX

A. SPECTROSCOPY OF CPVS OUTSIDE THE NARROW FLUX DIPS

In the main text we focus on photometric and spectroscopic signatures of the circumstellar material present around CPVs. However, due to the long-term unpredictability in CPV light curve morphology, some nights yielded coverage of only spot-induced or flare-induced broadband flux variability without providing any evidence—spectroscopic or photometric—for bound circumstellar material. We describe the results of those observations below.

A.1. *Keck/HIRES*

Figure 12 summarizes TESS and HIRES data for LP 12-502 on the nights of 23 November 2023, 3 December 2024, and 10 December 2024. The static spectra exhibit Balmer series emission up to n=15→2, strong chromospheric emission in the Ca II H&K lines, and the expected forest of molecular absorption lines for a non-accreting ("weak-lined") pre-main sequence M4.5Ve dwarf.

The HIRES spectra show flare-induced variability over all three nights. The most direct indicator for this variability is in chromospheric emission lines; excess emission appears during the variable epochs while the redder orders, including the order-normalized continuum, appear stable. This spectral variability is temporally coincident with elevated TESS flux levels (Figure 12), and is symmetric in the blue and red wings of the Balmer lines in the time-dynamic spectra, consistent with electric field-induced pressure broadening (A. F. Kowalski et al. 2017). This structure is mirrored across numerous chromospheric lines, including the entire Balmer series, the Ca II H&K lines, Ca I 4226, He I 4026, He I 4471, the Na doublet, and the Mg triplet. The flaring epochs also coincide with the excitation of at least a dozen Fe I lines spanning 3820 Å–4050 Å(A. Kramida et al. 2024). These lines are not present in the quiescent spectra; they appear in emission with $\Delta\lambda\approx0.2$ Å over the few-hour duration for which the other flare diagnostics are also present. This is consistent with established phenomenology for M dwarf flares (B. Fuhrmeister et al. 2008).

While the HIRES spectra and photometry of LP 12-502 presented abundant evidence for flare-induced variability, no variability could be clearly associated with any bound circumstellar hydrogen. Nonetheless, the same calculation as in Section 3.1 enables a conversion of this non-detection of Hα emission to an upper limit on the number density of bound circumstellar plasma. For LP 12-502, $L_\star$=0.011±0.001 $L_\odot$, $R_\star$=0.37±0.01 $R_\odot$ (L. G. Bouma et al. 2024), and we measure a mean stellar Hα EW and scatter of 0.22 ± 0.02 Å. The implied stellar Hα line luminosity is $\approx3.0\times10^{26}$ erg s^{-1}, two orders of magnitude lower than that of TIC 300651846. We derive an upper limit on the equivalent width of any excess Hα by injecting synthetic emission into the spectra shown in Figure 12. This process indicated that we would have been sensitive to circumstellar Hα emission with EW $\gtrsim$ 10 mÅ. As a result, we can rule out plasma clumps with number density $n_{\rm H}\gtrsim5\times10^{10}$ cm^{-3}, or assuming $r\approx0.1\,R_\star$, any clumps with $M_{\rm gas}\gtrsim6\times10^{15}$ g. This limit on the clump hydrogen number density is an order of magnitude lower than the densities observed for TIC 141146667 ($\approx1\times10^{11}$ cm^{-3}; L. G. Bouma & M. M. Jardine 2025) and TIC 300651846 (3–8$\times10^{11}$ cm^{-3}; Section 3.1).

A.2. *Hale/DBSP*

Figure 13 shows three near-adjacent nights observing LP 12-502 with Hale/DBSP. Figure 14 gives a complete visualization of the TESS flux and DBSP Hα EW behavior over the five adjacent nights in November 2024, folded on the 18.56 hour period. While LP 12-502 exhibited extreme variability from 2019–2023, showing four to eight sharp flux dips per cycle, it displayed only one or two sharp flux dips from May to December 2024 (TESS Sectors 78–86). A movie showing the star's TESS light curve from 2019–2025 is **available online**. In our November 2024 DBSP run, flare-induced variability played a significant role in modulating the Hα equivalent width measurements, and only two of the five nights (discussed in the main text) overlapped with the complex flux dips that define the CPV class.

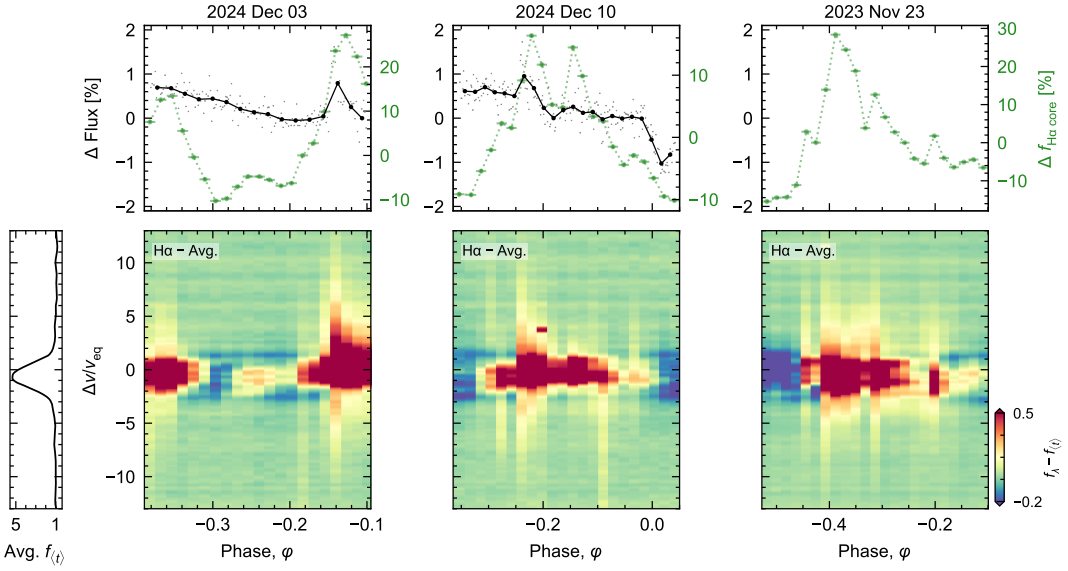


Figure 12. Keck/HIRES Hα time series of LP 12-502 vs. simultaneous TESS photometry. The lower left panel shows the time-average Hα profile on 3 December 2024. The successive columns show the time-dynamic mean-subtracted Hα profiles on the bottom. On top, simultaneous TESS flux measurements are in black, with the median-normalized Hα line flux at $|v/v_{\mathrm{eq}}| < 1$ in green. The variability on all three nights is flare-induced. The lack of (bound) circumstellar emission yields an upper limit on the number density of any plasma clumps, discussed in the text.

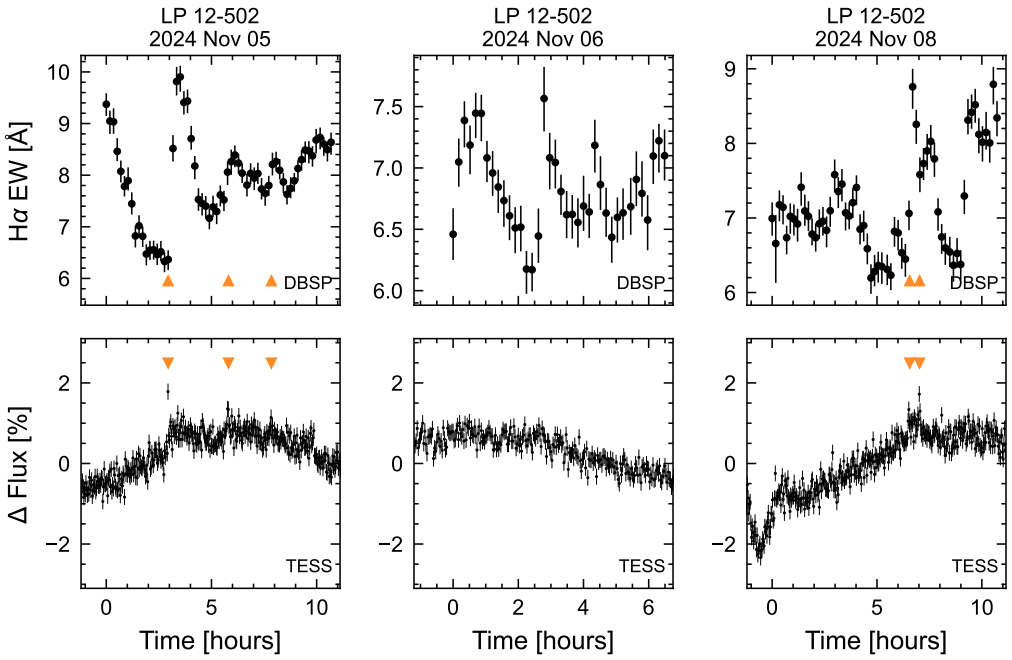


Figure 13. Hale/DBSP Hα equivalent widths of LP 12-502 vs. simultaneous TESS photometry outside complex dips. Panels are as in Figure 4, but for the three nights with only flare- and spot-induced variability. On Nov 5, flare-induced local maxima occur at t=3.5, 6, and 8 hours; on Nov 6 a similar maximum occurs at t=2.6 hours; and on Nov 8, at $t \approx 6.5$ hours. The Nov 8 DBSP observations began just after the end of a sharp "complex" flux dip.

B. TIME-SERIES PHOTOMETRY FITTING DIAGNOSTICS

Figures 15 and 16 show the results from our photometry fitting routine, which forms the basis for the chromaticity analysis in Section 3.3.

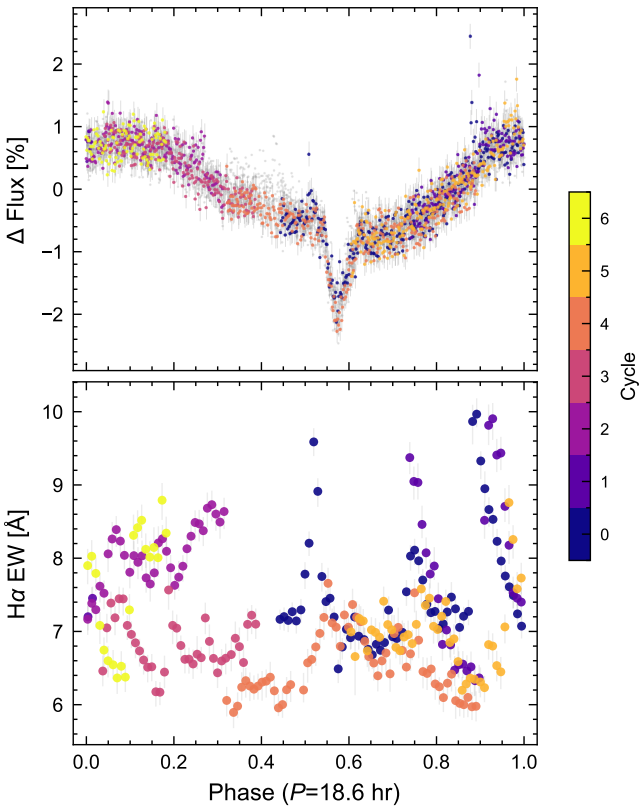


Figure 14. Hale/DBSP Hα equivalent widths of LP 12-502 vs. simultaneous TESS photometry, 4–8 November 2024. Points are colored by their orbital cycle number relative to the first 18.56 hour period. Simultaneous TESS measurements use the main colormap; TESS data in the 4 November to 8 November baseline that were not covered *simultaneously* by DBSP due to the day-night cycle are shown in gray.

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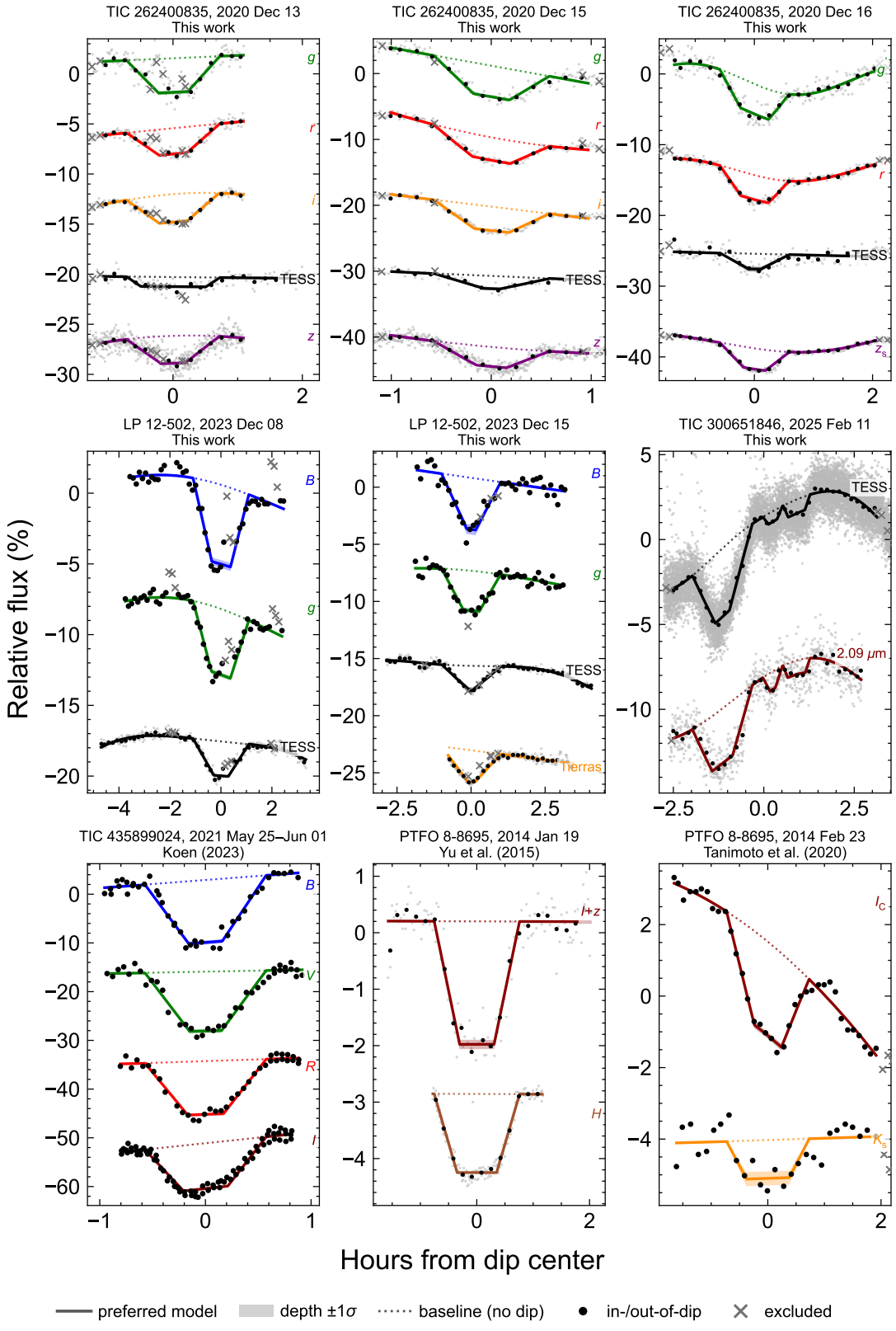


Figure 15. Trapezoidal fits to multi-band CPV photometry (1/2). Data from TIC 262400835, LP 12-502, and TIC 300651846 are new to this work. We digitally transcribed the measurements from TIC 435899024 (C. Koen 2023) and PTFO 8-8695 (L. Yu et al. 2015; Y. Tanimoto et al. 2020). Dip depth uncertainties (± 1-σ) are shown in the underplotted contours.

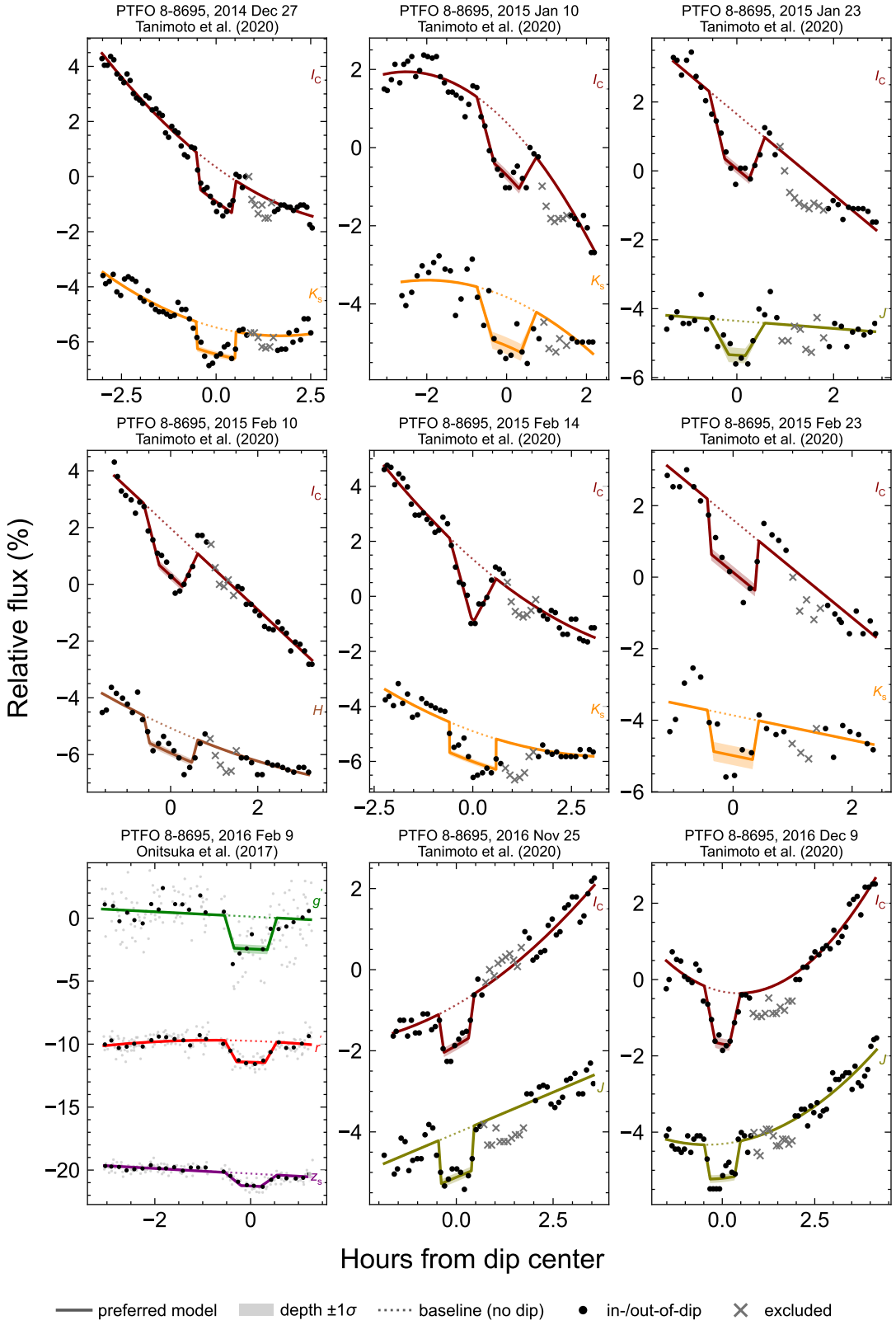


Figure 16. Trapezoidal fits to multi-band CPV photometry (2/2). We digitally transcribed these PTFO 8-8695 measurements from M. Onitsuka et al. (2017) and Y. Tanimoto et al. (2020). The underplotted contours show our fitted ±1-σ dip depth uncertainties.

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